Knowledge Work in the Age of LLM Agents:

Learning the Tacit Dimension of Expertise

Peking University Tsinghua University

Abstract. Large language models (LLMs) now pass professional licensing examinations and carry out well-specified office tasks, yet on assignments drawn from real practice they rarely deliver work that could be signed off as it stands. We take knowledge work broadly, as work in which knowledge is the principal means of production, and locate the main difficulty within it. Knowledge work is codified when its success criteria have been written down, as in factual knowledge QA and cowork tasks; codifiable when those who apply them could write them down, as with a team's workflow; and tacit when they rest on expert experience that those who apply them cannot state in advance. Each task passes through three stages—framing and planning, execution, and judgment and accountability—and the difficulty lies at the two ends of tacit knowledge work. AI research has concentrated on execution, and benchmarks usually supply the framing in the brief and stand in for judgment with a rubric written in advance. We review the evaluation, training, deployment, and societal impact of AI in knowledge work, and show that existing methods pass expert knowledge to AI mainly in written form, such as briefs and rubrics, which carry little of its tacit part. We expect execution to be increasingly automated and argue that framing and judgment should become collaborative between agents and professionals, a division of labor that sustains both progress in AI and the expertise on which it depends.

1 Introduction

"We can know more than we can tell."
— Michael Polanyi, *The Tacit Dimension* [1]

Large language models (LLMs) are extending from software engineering into knowledge work more broadly: they answer professional questions, carry out office tasks alongside their users, and increasingly contribute to the analysis, advice, and decisions on which organizations and individuals rely, in finance, law, medicine, scientific research, and beyond. This is where AI is expected to have its greatest economic and social impact. Research on AI in knowledge work, and on how it reshapes the professions, has grown rapidly but remains fragmented across domains and techniques.

Progress within knowledge work has been uneven. Models achieve passing-level scores on professional examination questions [2, 3] and approach industry experts in deliverable quality on well-specified occupational tasks [4], while APEX-Agents documents substantial delivery failures in simulated professional assignments under the model and tool configurations tested [5].

Full author list and contributions are given in Appendix A.



Figure 1. The three stages of knowledge work, illustrated with an equity-research case. Top: a codified proxy of the case, as typically set up for evaluating and training LLM agents; the brief frames the task and specifies the steps, a rubric written in advance stands in for judgment, and the agent carries out the analysis. Bottom: the same case in professional practice, where tacit knowledge surfaces at every stage, most heavily in framing and judgment: deciding what to ask, knowing where to look, and judging what the results mean before signing off. This survey examines how AI is evaluated, trained, and deployed across these stages, and how agents might learn what the bottom row requires.

Much of the difference lies in where a task begins and ends. An examination question or a detailed request hands the model a defined task and checks the result against a standard fixed in advance, whereas a real assignment begins with an open question and ends with a decision that must satisfy experts and for which someone is accountable. To explain this gap, we take knowledge work broadly, as work in which knowledge is the principal means of production [6], and locate the main difficulty within it. Knowledge work is *codified* when its success criteria have been written down, as in factual knowledge QA and cowork tasks; *codifiable* when those who apply them could write them down, as with a team's workflow; and *tacit* when they rest on expert experience that those who apply them cannot state in advance. The main difficulty lies at the boundary between codifiable and tacit knowledge work (Section 2).

Each task passes through three stages (Figure 1). In *framing and planning*, the worker decides what the problem is and how to approach it; in *execution*, sources are searched, analyses run, and drafts written; in *judgment and accountability*, the result is evaluated, the decision made, and someone signs for it. In codified and codifiable knowledge work, written or writable success criteria settle what the task is and what counts as acceptable, so the worker mainly executes. In tacit knowledge work, all three stages fall to the worker, and the two ends draw most heavily on tacit knowledge [1]: expert experience that experts act on but cannot readily state as general rules. Unlike the factual knowledge on which LLM research has concentrated, it shows itself only in particular cases. In the case of Figure 1, the analyst sees that a sharp rise in profit comes from a one-off sale of a plant, and she sees it from experience rather than from any rule. AI research has so far concentrated on execution. Across the 194 benchmark task subsets reviewed in Section 3, the brief sets the goal in every one, and the model chooses its own method in only 19%; framing is done by the task author, and judgment is usually replaced by a rubric written in advance.

For a long time, tacit knowledge kept computers out of professional work: tasks that could be written as rules were automated, and tasks that rested on tacit knowledge were not, a pattern Autor called Polanyi's paradox [7]. LLMs changed part of this. They learn from the large written record that experts have left, acquiring the knowledge in it without anyone stating rules for them. Research on that knowledge has extended from the facts a model stores or retrieves [8–10] to the

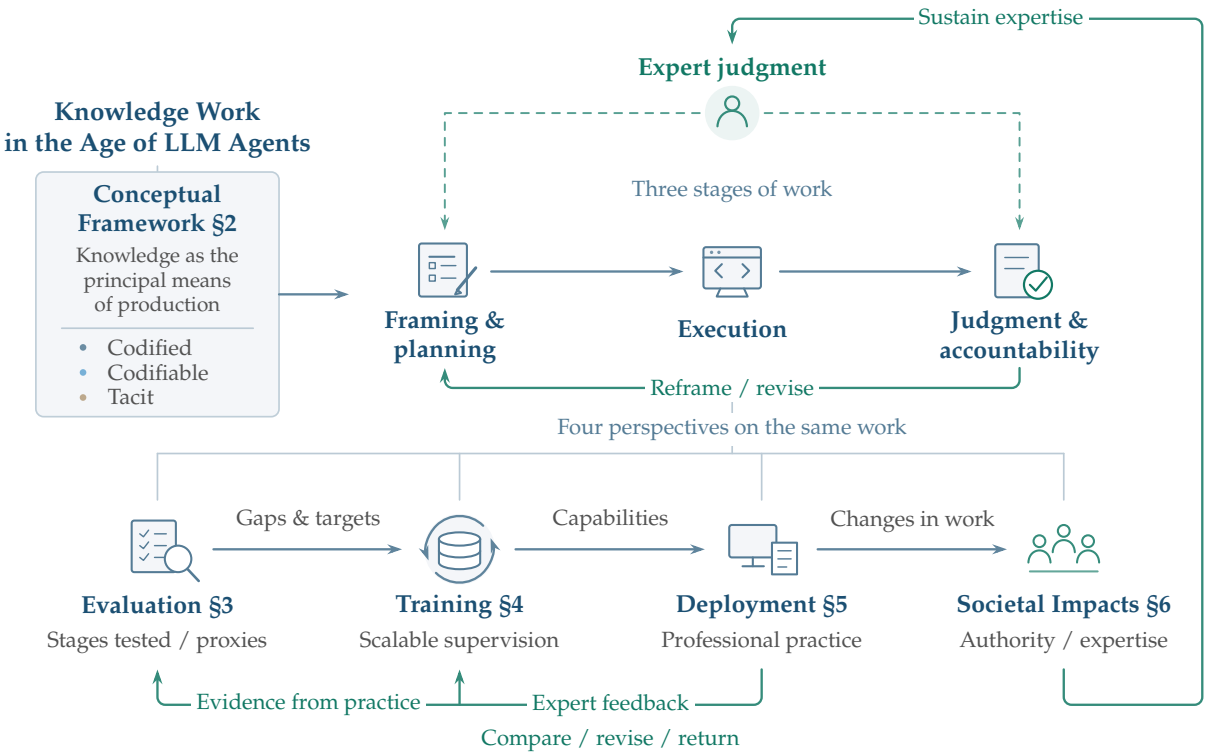


Figure 2. The survey's conceptual framework and the relations among its four themes. Section 2 defines the shared object, the three stages of knowledge work, in which expert judgment governs framing and acceptance. Evaluation identifies gaps, training builds capabilities, deployment returns evidence and expert feedback, and societal conditions shape whether expertise is sustained.

procedures an agent learns and follows [11–13]. LLMs now handle much codified knowledge work, codifiable work once its criteria are written down as procedures or skills, and many execution steps of tacit knowledge work. The success criteria at the two ends, however, still reach them in written form. Coding agents benefit from executable tests that can check specified requirements [14], but in investment research, law, medicine, and science such checks are rarely available: a brief can say "compute recurring profit growth" but not "this time, ask where the profit came from." Benchmarks let the brief do the framing and the rubric do the judging, and rewards derived from such rubrics exert little pull on what the rubrics leave out [15]. LLMs thus escape Polanyi's paradox in how they learn but meet it again in how they are supervised, a pattern we call the *second Polanyi paradox* (Section 2.1).

We argue that for agents to acquire the tacit knowledge at both ends, they need to learn from experts' actual work, for example which of two drafts an expert chooses, how she revises a clause, and why she returns a report to be redone. The professions have always passed on tacit standards through such acts, training newcomers through junior work done under supervision [16]. That junior work, however, is mostly codified or codifiable, the part that AI takes over first. When AI absorbs it, the professions lose a main route by which they train new experts, a loss we call the *periphery problem*; early evidence suggests that hiring of young workers has already slowed in AI-exposed occupations [17].

We take the following position. We expect execution to be increasingly automated, and we argue that framing and judgment should become collaborative: agents propose and draft, while professionals decide and answer for the result. This collaboration is sustainable only if authority stays with those who can check and answer for the work, expert choices are captured as learning signals, and training preserves the practice that automated execution removes.

This survey develops this position from capability to professional practice; Figure 2 shows

how its four core themes relate to the three stages of knowledge work. Section 2 defines knowledge work, the boundary between codifiable and tacit knowledge work, and the three stages. Section 3 asks which stages benchmarks test and shows how briefs and rubrics turn tacit knowledge work into codified proxies. Section 4 traces how environments, expert demonstrations, and expert feedback turn expert knowledge into learning signals, and how much of its tacit part they carry. Section 5 reviews how data, skills, tools, and specialized models supply factual and procedural knowledge at use time, leaving framing and judgment to professionals. Section 6 discusses how AI shifts authority and accountability in professional work, the training of new experts, and the distribution of costs and benefits.

2 Conceptual Framework

This section defines knowledge work, locates where within it LLMs fall short, and divides a task into three stages.

> **Definition (Knowledge Work).** Knowledge work is work in which knowledge is the principal means of production [6, 18].

Knowledge work ranges widely: from answering a professional question, through carrying out a specified task or an established workflow, to an open assignment that calls for expert judgment. It spans the professions and scientific research. We analyze it task by task, because technology typically substitutes for specific tasks within occupations [19] (Appendix B).

2.1 Tacit Knowledge Work

Knowledge work draws on three kinds of knowledge. Factual knowledge, the kind LLM research has largely studied, is what a model stores in its parameters [8, 9, 20] or retrieves from external sources on demand [10]. Procedural knowledge covers how work is done, such as a team's workflow [21, 22]. Expert experience is what practitioners accumulate in practice. These kinds differ in whether they have been written down and whether they could be [23]. Factual knowledge is largely codified, in statutes, guidelines, and textbooks. Much procedural knowledge is codifiable: unwritten, but those who follow it could write it down; Collins calls such knowledge relational [24]. Tacit knowledge is the part of expert experience that cannot readily be stated as general rules, even by those who act on it; it lies in perception, such as seeing what a case turns on, or in a community, such as what a court finds persuasive [1, 24]. Experts can state their judgment on a particular case, but not in advance the rule behind it [25]. Which kind of knowledge a task rests on shows in its success criteria.

> **Definition (Tacit Knowledge Work).** Tacit knowledge work is knowledge work whose success criteria cannot be stated in advance by those who apply them.

Success criteria cover what a task should accomplish and what counts as an acceptable result. Knowledge work whose success criteria have been stated in advance, as a question, a request, a reference answer, a test, or a checklist, is codified knowledge work; knowledge work whose success criteria have not been stated but could be by those who apply them is codifiable knowledge work (Figure 3). These distinctions are matters of degree and properties of the task itself.

Codified knowledge work includes factual knowledge QA, in which a defined question is checked against a reference answer [2, 3, 26], and cowork tasks, in which a request specifies a multi-step job on documents, data, or files [4]; rule-based processing is the most codified case.

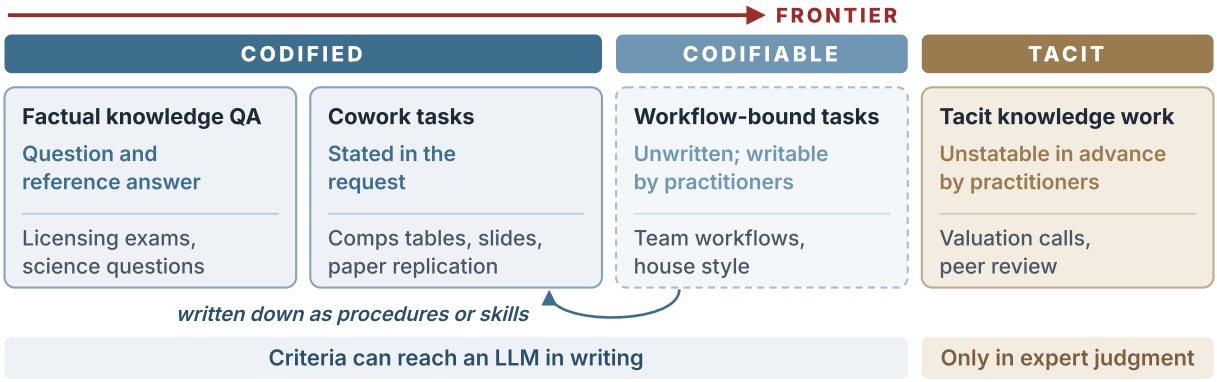


Figure 3. Knowledge work by the state of its success criteria. Each card gives the criteria (top) and examples (bottom). Writing codifiable criteria down as procedures or skills moves the work to the codified side. The frontier of AI has crossed codified work into codifiable work; tacit work lies beyond it.

Codifiable knowledge work follows a team's workflow, a firm's house style, or a reviewer's checklist. An LLM given only the request lacks this context, and writing the criteria down as a procedure, a template, or an agent skill supplies it (Sections 5 and 6). Tacit knowledge work, such as a light-brief Harvey LAB matter [27], leaves the worker to decide what matters, with no reference that valid outputs converge on; here LLMs still fall short. Research spans the same range: graduate-level science questions [28]; replication, where the paper serves as both brief and rubric [29]; a lab's protocols; and choosing questions and reviewing manuscripts.

The main difficulty for AI in knowledge work therefore lies at the boundary between codifiable and tacit knowledge work. The boundary moves: LLMs learn from what experts have left in writing, and organizations write down what was codifiable, so work shifts toward the codified side; what cannot be stated moves slowly. The boundary has two further consequences that recur throughout this survey, one for how AI is supervised and one for how experts are trained.

First, LLMs can learn a task without experts stating its rules, yet the success criteria for their work still reach them mainly in forms written down in advance: a brief, a rubric, a reward, or a skill (Sections 3 to 5). These forms carry codified and codifiable criteria well and tacit criteria poorly. We call this the *second Polanyi paradox*: Polanyi's paradox moves from how machines perform work to how they are supervised. Generative AI engages mainly the expressed outcomes of tacit knowledge [30], and knowledge an LLM extracts from experts' observed tutoring helps novices more than the experts' own account of it [31]. Expert acts of comparing, revising, and returning work can therefore carry what written criteria leave out (Section 2.2).

Second, tacit knowledge is acquired in practice. Newcomers begin with legitimate peripheral participation, taking on simple, low-risk parts of real work under supervision [32], such as document review, first drafts, comparable-company tables, and clinical notes; in research, graduate students start with literature reviews, data processing, and replications. Most of this is codified or codifiable knowledge work, and the properties that make it suitable for newcomers make it the easiest to hand to AI. The work through which newcomers acquire the tacit knowledge of their field is thus the work AI absorbs first, a loss of training we call the *periphery problem* (Section 6.2).

2.2 Three Stages

A knowledge-work task passes through three stages. *Framing and planning* turns an open question into a defined task. *Execution* carries out the task, producing results that can be checked against it. *Judgment and accountability* evaluates the results, makes the decision, and answers for it. The stages are logical rather than chronological; execution often sends a task back to fram-

ing (Figure 1). In terms of Bloom's taxonomy of cognitive processes [22], they are chiefly *Create*; *Understand*, *Apply*, and *Analyze*; and *Evaluate*, respectively. Success criteria govern the two ends: what the task should accomplish is settled in framing, and what counts as acceptable is applied in judgment. In codified and codifiable knowledge work, written or writable criteria settle both ends, so the worker mainly executes; tacit knowledge work leaves both ends to the worker's judgment.

Framing and Planning. Professional problems do not arrive defined; they "must be constructed from the materials of problem situations which are puzzling, troubling, and uncertain" [33], and those materials often include context no textbook contains: a client's history, a firm's conventions, the tendencies of a court. LLMs readily recall what might be relevant; the hard part is deciding what is relevant, what should be delivered, and where to begin.

Execution. Execution means searching sources, extracting data, computing measures, and drafting documents. Most of its results can be checked: whether a figure can be recomputed from the original filing, whether a citation is accurate, whether a format complies. In professional organizations, execution is largely done by junior staff and reviewed by seniors, and it is the stage in which newcomers start. It also takes experience, as when an analyst who sees profit rise while cash flow stalls knows to read the note on non-recurring items; because such experience can be checked against results, it is the easiest to learn from records of practice.

Judgment and Accountability. The results of execution become a work product only after they are evaluated. Experts judge whether the work is sound and which option suits the case, and decide accordingly, as when an investment committee trims a position, a physician signs an order, or an auditor issues an opinion; whoever signs is accountable for the result. The standards at this stage are often hard to state and outcomes often arrive late, so the professions substitute peer judgment for outcomes not yet observed, as science substitutes peer review for a verdict that the research community reaches only over time.

Judgment is rarely exercised by one person. Professional work is organized in delegation chains, such as partner, associate, and paralegal. In Fama and Jensen's terms, juniors exercise *decision management*, initiating and implementing, while seniors exercise *decision control*, ratifying and monitoring [34]. Aghion and Tirole further distinguish *formal authority*, the right to decide, from *real authority*, which accrues to whoever holds the relevant information [35]. AI enters these chains near the bottom, so acceptance matters more. If professionals cannot check what an agent produces, real authority shifts to the agent while accountability stays with the person who signs, a condition we call *authority drift*. Acceptance also passes on tacit standards: when a senior accepts, revises, or returns a draft, or an adviser comments on a student's manuscript, the decision carries standards the reviewer may never state. A professional reviewing an agent's output makes the same decisions, so these decisions can also become a source from which agents learn tacit knowledge.

3 Evaluation

Knowledge-work deliverables rarely have a single correct answer, so a benchmark score means only as much as two design choices allow: what the benchmark leaves to the model, and how it checks the result [36]. Coding 194 task subsets from 44 benchmarks, we find that most supply framing and planning in the brief and test execution within that frame, and that newer benchmarks that delegate more depend on graders whose validity is unestablished. Both patterns

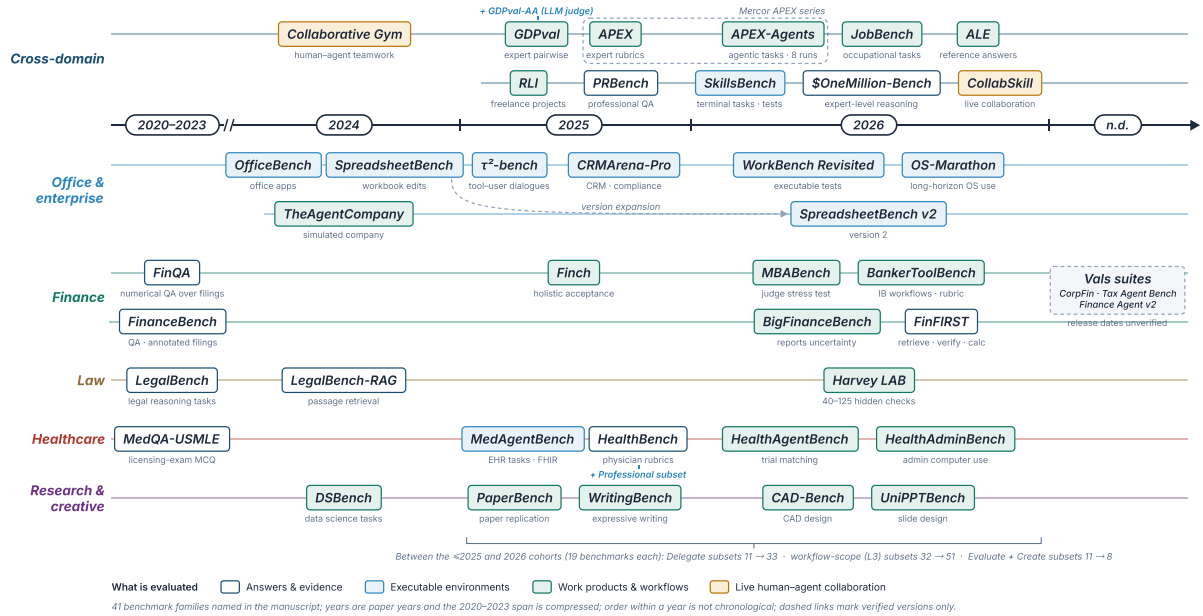


Figure 4. Timeline of benchmark families discussed in this survey. Fill colors group benchmarks by their primary evaluation focus, rather than the formal cognitive and automation levels. The 41 families shown differ from the 44-benchmark statistical catalog; the cohort comparison refers to the 38 dated benchmarks in that catalog.

Dimension	Levels	Coding criterion	Theoretical anchor
1 Cognitive level	Understand · Apply · Analyze · Evaluate · Create	Where the main difficulty lies, not how complete the deliverable is.	Revised Bloom taxonomy [22]
2 Automation level	Assist · Collaborate · Delegate · Autonomous	Who decides the goal, materials, method, structure, and substantive conclusion in the brief the AI receives.	Decision management vs. control [34]; formal vs. real authority [35]

Table 1. Coding framework for knowledge-work benchmarks. Each task subset receives one level on each axis, and the two axes define the cross-tabulations used throughout this section. Levels are shaded light to dark.

follow from what can be graded: only what can be written down in advance, as a reference, an executable check, or a rubric. A benchmark thus creates a codified proxy for tacit knowledge work and leaves unmeasured the reaction of the expert who would sign the work (Section 2.1).

3.1 Analytical Framework

A single benchmark often bundles heterogeneous tasks under one domain label, so we code each benchmark's published *task subsets* on two primary axes (Table 1) and record two auxiliary annotations, task scope and scoring mode.

Cognitive Level. Following the revised Bloom taxonomy introduced in Section 2.2 [22], the first axis records the operation that creates a task's main difficulty: extracting explicit information (*Understand*), carrying out established procedures toward a determinate result (*Apply*), decomposing a problem and reconciling evidence (*Analyze*), judging quality against a standard (*Evaluate*), or originating structure and substance where many outputs are acceptable (*Create*). In professional work, *Create* is constrained design rather than artistic expression: devising an argu-

	Automation level						Scoring mode			
Cognitive level	**Assist**	**Collab.**	**Deleg.**	**Auton.**	**Total**	**% Del.**	**Determ.**	**Rubric**	**Human**	**% Det.**
Create	0	5	**2**	0	**7**	29%	0	4	3	**0%**
Evaluate	7	2	**5**	0	**14**	**36%**	8	6	0	57%
Analyze	14	36	22	1	**73**	30%	30	36	7	41%
Apply	18	**51**	13	2	**84**	15%	**51**	24	9	61%
Understand	11	1	4	0	**16**	25%	11	5	0	69%
Total	**50**	**95**	**46**	**3**	**194**	24%	**100**	**75**	**19**	52%
% Determ.	60%	48%	46%	100%						

Table 2. Task subsets by cognitive level (rows, highest first) against automation level (left) and scoring mode (right); 194 subsets from 44 benchmarks, counted per subset. Shading scales with the count. Teal cells mark the upper-right region, *Evaluate* or *Create* work under *Delegate* or *Autonomous* operation. Gray columns give row shares; the bottom row gives the deterministic share by automation level. Collab. = Collaborate; Deleg. = Delegate; Auton. = Autonomous; Determ. = deterministic verification; Rubric = rubric-based grading; Human = human evaluation.

ment, transaction structure, treatment plan, or policy when existing rules and evidence do not single one out. The axis thus roughly indicates which stage of a case a task tests, though not one to one: an expressive *Create* task done under a given brief is still execution, and checking a result against a stated rule is *Evaluate* in Bloom's terms but execution in ours. Cognitive level is independent of *task scope*, which runs from a single step (L1) to an ongoing function (L4): drafting a contract from prescribed clauses yields a complete file but is *Apply*.

Automation Level. The second axis records how much decision authority the brief hands to the AI, coded from the brief as the AI receives it: who decides the *goal*, the *materials* that matter, the *method*, the *structure* of the deliverable, and the *substantive conclusion*. The first three are decided in framing and planning; the conclusion is reached within the frame and differs from the judgment stage, in which a result is evaluated and accepted. The goal is the principal's by construction in any benchmark, so the levels are separated by the other four decisions and by whether the result is ratified. Rubrics hidden from the model are not part of the brief (Appendix C). The levels follow the division between decision management and decision control (Section 2.2). In **Assist**, the AI implements a single step and the human decides what happens next; in **Collaborate**, it implements what the principal has initiated, with the method or conclusions already fixed; in **Delegate**, it also decides which materials matter, how to proceed, and what to conclude before a human ratifies the result; in **Autonomous** operation, nothing is ratified. A score under a heavy brief thus shows whether a model can carry out a specified approach, and a score under a light brief whether the work can be handed over.

3.2 Benchmark Landscape

The catalog spans finance, law, medicine, science, education, engineering, management, office administration, creative production, public services, and customer operations, excluding software development and cybersecurity (Table 2). Software is excluded as a sample and used as a reference point (Appendix B): its process is recorded in the repository and its output can be executed, so its benchmarks can check outcomes directly, and the patterns below show what happens where neither holds.

Heavy Briefs Dominate. Most subsets are *Apply* or *Analyze* under *Collaborate*: suites present complete professional assignments while the prompt fixes the formulas, parameters, layout, or even the conclusion. Even GDPval [4], built from occupational work products, is mainly *Apply* in seven of its nine sectors; one government task asks for a revised staffing plan but supplies the reduction percentage and the positions lost, so the AI carries out a decision already made. A Harvey LAB task of similar scope gives a one-sentence objective and a folder of matter documents, leaving what to look for and what to conclude to the AI [27]. Across the five coded decisions, the principal sets the goal in all 194 subsets, and the AI reaches the substantive conclusion in 122 (63%) but chooses the materials that matter in only 27 (14%) and the method in only 36 (19%). Benchmarks thus test conclusions reached within a given frame far more often than framing and planning, the first stage of a case (Section 2.2).

Setting Outweighs Domain. Within one domain, benchmarks span the automation scale. In finance, CorpFin poses single questions over filings (*Assist*), the investment-banking track of APEX-Agents prescribes the computation (*Collaborate*), and Tax Agent Bench leaves the method to the AI (*Delegate*) [5, 26]; law and medicine show the same spread [27, 37, 38]. A claim that models score x on legal work is incomplete until it states how much of the work was handed over.

Delegated Judgment and Creation Are Rare. Delegated *Evaluate* work is measured mainly when judgment ends in a checkable decision, such as clinical-trial matching in HealthAgentBench or policy-compliance checking in CRMArena-Pro [39, 40]. All seven *Create* subsets test expressive writing, visual design, or slide layout [41–44]; substantive creation in law, finance, medicine, and policy goes untested.

Takeaway: Benchmarks Test Execution Within a Frame and Miss Professional Creation

Most subsets test *Apply* or *Analyze* under briefs that already fix the method or the conclusions. Delegated *Evaluate* work is measured mainly where judgment ends in a checkable decision, and the few *Create* subsets test expressive writing or visual design rather than the constrained design of professional work—devising a transaction structure, a litigation strategy, or a treatment plan.

3.3 Task Construction

Heavy briefs dominate mainly because of what benchmark authors can grade. Some examples in this subsection and the next come from outside our catalog.

Grading Shapes the Brief. Ouchi's account of organizational control [45] explains the pattern. Organizations control work by specifying behavior when the process is well understood, by measuring output when results are observable, and by relying on professional norms when neither holds. Benchmark authors face the same choice. *Collaborate* subsets control behavior: the brief fixes the method, so any competent execution converges on the reference. Verifiable *Delegate* subsets control output: the method is open but the result can be checked, as in OS-Marathon [46] or DSBench [47]. Where quality calls for professional judgment, benchmarks fall back on rubrics scored largely by LLM judges, asking a written checklist to stand in for professional norms. Accordingly, deterministic checks become rarer as cognitive level rises but barely change with automation level (Table 2).

Construction pathways reinforce the pattern (Table 3). Whether a task adapts existing materials, follows an expert's design, or reconstructs past work, it needs a reference that valid outputs

Pathway	Starting point	What outputs must converge on	Examples
Adaptation	Existing documents, records, or research artifacts, turned into newly written requests.	An answer derivable from the source; often secured by writing the method into the brief.	FinanceBench† on company filings [49]; PaperBench on published papers (*Delegate*) [29]
Expert design	Professional experience and target capabilities, translated into scenarios and standards.	An expert-written rubric or reference deliverable.	APEX-Agents (*mostly Collaborate*) [5]; GDPval (*7 of 9 Collaborate*) [4]
Reconstruction	Past requests, communications, file changes, and deliverables.	The historical deliverable, or whether a client would accept the output.	RLI paid projects (*6 of 7 Collaborate*) [42]; Finch† email-to-artifact workflows [50]

Table 3. Three task-construction pathways, distinguished by their starting point. Pathways can coexist within one task. Automation levels in parentheses are from our coding; † marks benchmarks outside the 44-benchmark catalog.

converge on. Suites that require a reference answer exclude open-ended tasks outright, as ALE does with design tasks [48], and the simplest way to make outputs converge is to write the steps, assumptions, and schema into the prompt, turning a delegated task into a collaborative one. Delegation survives where the source fixes the goal and leaves the route open, as when PaperBench asks for a published paper to be replicated [29].

Context Stays Private. Tasks must be built from releasable materials, so the institutional context practitioners accumulate, such as client history and internal precedent, is largely absent, even though much of it is codifiable (Section 2.1); a real engagement builds context gradually, while a benchmark hands it over as a folder. The same confidentiality keeps the expert reactions to that work, the redlines and returned drafts of Section 2.2, out of the record from which benchmarks and training data are built. Realism also varies within a task, as when a genuine filing supports a newly written prompt in a synthetic company [51], so claims of realism should address the request, inputs, environment, and reference separately.

Heavy Briefs Mirror Junior Work. Much entry-level work is done under exactly such briefs: a junior analyst extends a model whose structure a senior has set, and a junior associate drafts from an agreed term sheet. Benchmarks concentrated in *Apply* and *Analyze* under *Collaborate* may represent junior work faithfully while saying little about framing problems and answering for the result. This is the periphery problem of Section 2.1 seen from the side of evaluation, consistent with tentative evidence of slower hiring of young workers into AI-exposed occupations [17].

Takeaway: To Be Gradable, Benchmarks Fix the Method

To make outputs gradable, authors write the method into the brief or admit only tasks with checkable results; where neither works, rubrics stand in for professional judgment. Each option measures against something recorded in advance, so none measures whether the expert who would sign the work would accept it. The resulting tasks resemble junior work done under a senior's brief, stripped of the institutional context in which real work is done.

3.4 Scoring and Calibration

Once a task is fixed, its score depends on the basis against which outputs are compared, whether the model can see that basis, and how criteria are combined.

Scores Certify Less Than Completion. Each common basis certifies something narrower than task completion. A reference answer certifies the endpoint but not how sources were chosen [52]; an executable test certifies only the properties it asserts [53]; a pairwise comparison certifies preference without confirming that critical requirements were met, so GDPval's expert comparison and GDPval-AA's LLM judge give the same tasks different meanings [4, 54]; and a rubric certifies the criteria it lists. Holistic acceptance, such as Finch's check that a workflow finished without critical errors [50], comes closest to a principal's sign-off. Expert comparisons and expert-written rubrics are also where professional verdicts on AI work are now recorded at the largest scale; they are used to score, and Section 4.3 asks what it would take to learn from them.

Hidden Rubrics Retain Authority. A Harvey LAB task gives the model a short instruction but scores the output on 40 to 125 binary checks the model never sees [27, 55]. The checks are a lawyer's verdict written down before the work is done. Formal authority has passed to the AI, while real authority, held by whoever has the information that defines success, stays with the benchmark author; this reverses the drift of Section 2.2 because the author, unlike a professional reviewing an agent in practice, holds information the agent lacks. Anticipating unstated expectations is a professional skill, but such a score also rewards agreement with one author's preferences among defensible alternatives.

Who Catches Errors Decides Which Score Matters. Aggregation rules encode different notions of success: the share of criteria satisfied measures **artifact quality** (Q) [56], passing every necessary criterion in one run measures **complete delivery** (D), and passing across repeated runs measures **sustained reliability** (R); **real-world uptake** (U) comes from usage data [57, 58]. They can differ several-fold for one system: in the evaluation reported in APEX-Agents v3, Gemini 3 Flash scores 39.5% on Q, 24.0% on D, and 13.4% when all eight runs must succeed [5]. Which one matters depends on who reviews the output. Under *Assist*, a human checks each step and can repair what is missing, so Q is useful; under *Delegate*, the human sees only the finished deliverable, so D is the question; under *Autonomous* operation nobody reviews, and R becomes the requirement. Because one material error can make an otherwise strong memo unusable, rubrics for delegated tasks should separate disqualifying requirements from credit-bearing ones, and reports should give uncertainty estimates under stated tool, prompt, budget, and retry conditions [59].

Judging Open-Ended Work Involves Three Questions. Work without a reference answer is often called subjective, but professional quality is not a matter of taste: peers share standards they cannot fully state (Section 2.2). Judging an output involves three questions: does it contain a disqualifying error, would an expert sign it as delivered, and which acceptable output suits this client better? Experts can be expected to agree most on the first and least on the third, where disagreement often reflects legitimate alternatives rather than noise. Rubric scores and pairwise comparisons mix the three. Evaluations should check errors automatically where possible, put acceptance to experts as a sign-off decision, and report preferences as context-dependent, together with expert agreement.

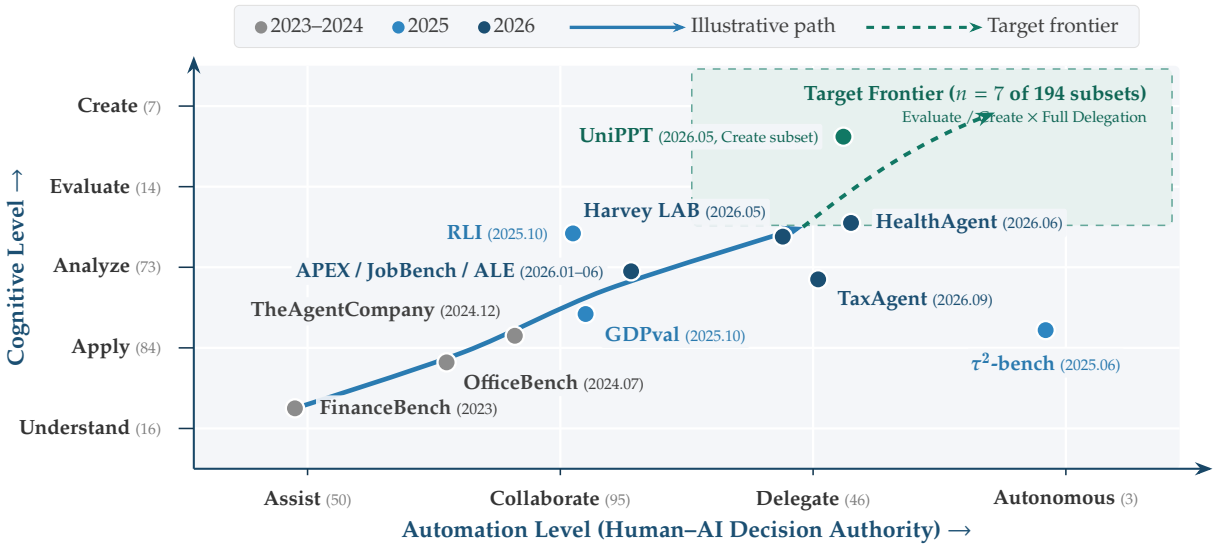


Figure 5. Illustrative placement of selected benchmark task settings across automation level (horizontal) and cognitive level (vertical), colored by release cohort. Positions and connecting paths are schematic, not benchmark-wide scores or evidence of a monotonic upward trajectory. Coding is performed at the task-subset level. The shaded region contains 7 of 194 subsets assigned to *Evaluate* or *Create* under *Delegate* or *Autonomous*; this count does not establish coverage of open-ended professional judgment or design.

Judges Need Expert Calibration. Validity is what rubric-and-judge pipelines most need. Recent work pursues it in construction, as when APEX-Agents corrects rubrics through expert attempts [5]; in grading, as when EvalGen selects evaluators by agreement with expert grades [60]; and in stress testing, as when MBABench [61], Mubric [62], and *A Judge Should Know What Changed* [63] check whether judges detect injected errors and substantive edits. The reviewed studies do not yet establish general validity for open-ended professional work. Each is an instance of the same procedure, an expert judging sampled outputs and the criteria being revised, that Section 4.2 describes for training data.

3.5 Trends and Open Gaps

We split the 38 dated benchmarks by first release into 19 from 2025 or earlier and 19 from 2026. The cohorts are not matched on domain, so the comparison is descriptive. Figure 5 maps selected benchmark task settings by cohort.

Delegation Stops at Analysis. *Delegate* subsets rise from 11 to 33, while *Evaluate* and *Create* fall from 11 subsets to 8. More delegation has so far meant handing over analytical method, not open-ended judgment or design. The two *Create* subsets added in 2026, in JobBench and UniPPTBench [43, 44], again concern artistic and slide design, and no benchmark in either cohort asks a model to devise a transaction structure, a litigation strategy, or a treatment plan.

Scope Grows Within Single Episodes. Workflows (L3) grow from 32 to 51 subsets, so tasks increasingly span many steps. They remain bounded episodes, however: the brief and materials arrive at once and the task ends with a deliverable. Continuous responsibility for an organizational function (L4) is nearly absent, and live collaboration with real people appears only in Collaborative Gym [64] and CollabSkill [65]; real-world uptake (U) is accordingly measured only outside benchmarks, in usage data [57, 58]. Professional work on a case unfolds over time, as context accumulates and the client responds; benchmarks have lengthened the episode without yet capturing this.

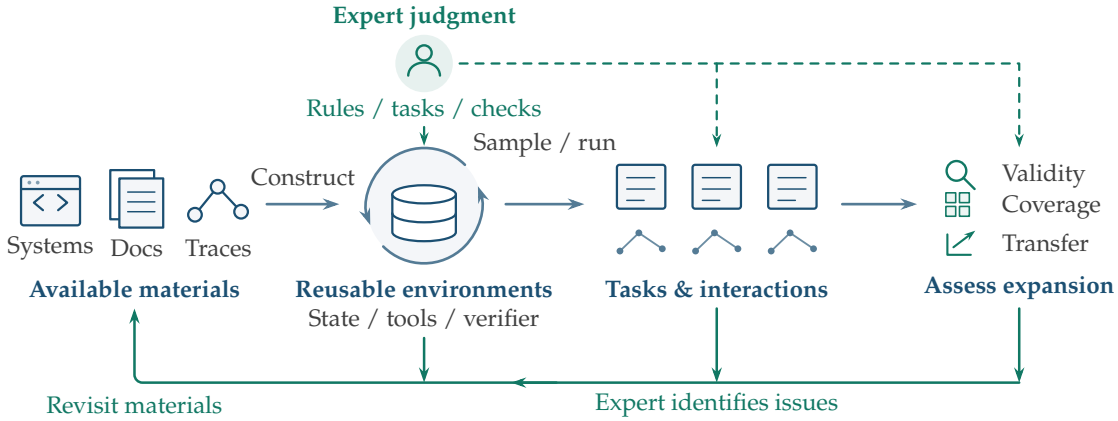


Figure 6. A conceptual workflow for scaling training supervision, illustrated through environment construction and experience generation. The shared principle is to turn available professional materials into reusable supervision under expert guidance. Experts oversee construction, generation, and assessment; issues they identify at any stage prompt a return to the materials for revision and regeneration. Not all surveyed systems implement this complete loop.

These findings carry over to training and deployment. The verifiers and rubrics built for evaluation become rewards in reinforcement fine-tuning, so benchmarks dominated by heavy briefs reward executing a given method, and the calibration problem of Section 3.4 becomes a reward-hacking problem once the judge supplies the gradient. The automation level coded here describes benchmark design, not work arrangements in practice; Section 5 examines the knowledge, procedures, tools, and operating conditions needed to bring model capabilities into professional use.

4 Training

Training for knowledge work should support all three stages: framing and planning, execution, and judgment and accountability. A brief request may omit goals, constraints, and context that professionals uncover through interaction and experience. Models can miss consequential gaps in legal queries or answer underspecified requests without eliciting users' preferences [66, 67]. Professional framing therefore still calls for expert involvement in deciding what the request requires and translating it into an actionable task. In many training settings reviewed here, this work is performed upfront: experts help design tasks and specify requirements, procedures, and acceptance criteria, leaving the policy to practice within an established frame. Such settings underrepresent the discussion through which a task is clarified and revised. User-agent simulation begins to address this gap: PersonaForge synthesizes multi-turn training interactions in which simulated users progressively disclose requirements and correct intermediate work [68], while IntentRL trains agents to clarify user intent before research through offline dialogues and online interaction with a user simulator [69]. These approaches expand interaction experience, although transfer to situated professional judgment still requires validation. Against this background, we examine how limited expert input becomes supervision at scale: environments expand opportunities for practice, synthetic reasoning expands demonstrations, and rubric synthesis expands criteria for training feedback. These are complementary routes to supervised fine-tuning (SFT) and reinforcement learning (RL), whose coverage of the three stages depends on which decisions remain open to the model. Scale should be assessed through added valid experience and gains on unseen work. Further analyses and case studies are provided in Appendix D.

4.1 Scaling Training Environments

Expert Judgment Enters Through Environment Design. Expert judgment enters a training environment through decisions about the work, the actions available, and the conditions for completion. Professional procedures and constraints can be encoded in initial states, tool behavior, task instructions, and verifiers. This makes those judgments reusable across training interactions without requiring an expert to supervise every attempt. Such checks capture specified aspects of professional quality; their adequacy remains a question for validation. Scaling then means expanding the tasks and situations in which the policy can practice under these constraints. A remaining gap in framing and planning is that current scaling methods largely expand tasks whose requirements have already been specified with expert input, whereas real-world requirements may emerge and change through human–AI dialogue [68].

Constructing Environments at Scale. Coding research has developed methods for synthesizing training tasks at scale from executable repositories, providing construction strategies that may be adapted to professional workflows [14, 70]. Other approaches construct the environment itself: ScaleEnv and AutoForge generate tools, databases, and tasks from domain descriptions or tool documentation, using execution to check reference solutions [71, 72]. DreamGym instead learns an experience model from offline trajectories to predict subsequent states and rewards, reducing the need for new online execution [73]. These routes turn existing systems, documents, or interaction records into reusable sources of training experience. Their validation requirements differ: generated systems need consistent execution rules, while learned simulators also need evidence of transfer to real execution.

Professional Workflow Examples. EnterpriseLab builds dependency graphs over enterprise tools, samples executable paths, and generates user intents for training [74]. Corecraft assembles linked orders, inventory, policies, and tickets into a coherent organization, then uses expert-authored tasks and rubrics for RL [75]. HealthAdminBench provides healthcare-administration workflows across electronic health records, payer portals, and fax systems, with a small SFT experiment demonstrating adaptation to the tested workflow [76]. Spreadsheet-RL converts forum problems into initial and target workbooks and obtains execution feedback by recalculating edits [77]. BankerToolBench extends tasks to interdependent financial models, presentations, and memos, and reports offline post-training on frozen rollout pools [78]. Together, these cases show how professional materials and procedures can support training, although they do not establish a common scaling law.

What Counts as Effective Expansion. More instances do not necessarily add useful experience: EnterpriseLab reports non-monotonic gains as training-set size increases in a fixed setting [74]. Effective expansion should add relevant states, constraints, and failure modes while keeping instructions, environment behavior, and verifiers consistent. It should also produce feedback that distinguishes behaviors the current policy can generate. In the proposed cycle, experts review construction, generated experience, and assessment. Issues identified by experts at any stage can send the process back to source materials for correction or supplementation before reconstruction and regeneration. We therefore favor reporting validated interactions, task and state coverage, collection and expert-review costs, and independently assessed performance on unseen work alongside task counts. This separates the ability to generate more experience from evidence that the added experience improves professional work.

4.2 Synthesizing Expert Reasoning

Finished professional work often records a decision without the reasoning behind it. Synthetic chain-of-thought (CoT) uses models to produce candidate reasoning from available cases and professional knowledge. Instead of asking experts to author every trace, synthesis can reuse their questions, procedures, examples, and judgments across many generated instances. We organize the evidence around four stages: seeding, generation and filtering, expert feedback, and expansion. The reviewed studies implement these stages to different degrees, and we distinguish reported mechanisms from the proposed feedback cycle.

Seeding Generation with Professional Knowledge. The seed supplies content and constraints that guide generation beyond a generic request to explain an answer. Fin-o1 uses subquestions collected from financial experts and program steps; LegalOne combines expert protocols and knowledge bases in a workflow of fact finding, issue identification, rule retrieval, and derivation [79, 80]. MedReason anchors generation to paths through a medical knowledge graph [81]. Expert elicitation provides another source: BRIDGE turns tutors' explanations of what they noticed and why they responded into a decision structure [82]. These inputs differ in how directly experts authored or inspected them, but each can guide many generated instances. The opportunity for scale is to reuse professional constraints across cases, reducing the need for a newly written expert rationale for every example.

Generation and Automatic Filtering. STaR iterates generation, filtering by answer correctness, and fine-tuning [83]. Professional pipelines add revision: Agentar-DeepFinance uses reference answers to diagnose failed reasoning, generate repairs, and verify them; MedReason retains chains from which a model recovers the reference answer [81, 84]. Generation and automatic checking expand the candidate pool without requiring a professional to inspect every attempt. However, endpoint correctness does not establish that the reasoning used appropriate evidence or conditions, and a generator and verifier can share blind spots. Information available during construction must also be distinguished from information available to the deployed student. Reference answers may guide synthesis, but the demonstrated steps must remain executable from the evidence the student will have.

Turning Expert Review into Revision and Supervision. Expert review can test whether plausible traces reflect acceptable professional reasoning. MedReason reports physicians' preferences for sampled chains, but uses them as a one-shot quality assessment rather than feedback to its generator [81]. This distinguishes collecting a verdict from using it to improve subsequent training. Feedback has two possible routes. First, an expert can identify a missing condition, faulty assumption, or inappropriate procedure; that correction can revise the seed, add a contrasting example, or trigger regeneration of affected cases. Correcting the generating rule can propagate a lesson beyond the reviewed sample. Second, comparisons, edits, or process labels can directly supervise a policy or evaluator, even when the expert cannot provide a complete verbal rule. Process supervision and imitation learning from language feedback provide precedents outside the professional settings considered here [85, 86]. The reviewed professional studies provide stronger evidence for seeding and automatic filtering than for repeated expert-feedback cycles. Repeated expert-feedback cycles therefore require further testing in professional training, with their actual use distinguished from final evaluation.

Expanding Coverage and Testing Training Value. Useful expansion should add valid examples of relevant cases, boundary conditions, and failure modes. We propose combining targeted

review, which seeks informative failures, with random auditing, which estimates rejection under a specified review protocol. Rejections should lead to revision or regeneration before further expansion; acceptance of a sample does not establish that every generated trace is sound. Ultimately, validation must test whether the resulting data improve independently assessed work on unseen cases. Fluent rationales need not reveal the computation that produced an answer [87]. LegalOne, Fin-o1, and MedReason report gains from combined training recipes, so their results do not isolate the contribution of expert feedback. Coverage, review cost, and downstream performance together determine whether synthesis has converted limited expert input into useful training supervision.

4.3 Synthesizing Rubrics for Scalable Supervision

Rubrics turn requirements for acceptable work into explicit criteria. Evaluation asks what scores establish about capability; training asks how to produce enough useful criteria and feedback to improve a policy. The scaling opportunity is to have models synthesize task-specific rubrics from available professional materials and expert guidance, then reuse them across training responses. This expands both the supply of rubrics and their application through automated scoring.

Generating Rubrics in Batches. Rubrics as Rewards (RaR) uses an LLM to generate prompt-specific criteria from reference answers in medicine and science, treating these answers as proxies for expert supervision. Criteria distinguish essential requirements, optional improvements, and pitfalls [88]. ARES extends synthesis to raw documents: a model jointly generates question–answer pairs and weighted rubrics, with automatic checks for question self-containment, answer faithfulness, and rubric validity. It constructs 100,000 instances across ten domains [89]. These approaches translate existing knowledge into many reusable scoring specifications. Expert input can thus enter through source materials, reference answers, and generation principles; neither approach requires an expert to author each rubric.

Reusing Criteria as Training Rewards. Once generated, a rubric conditions an automated judge that scores successive policy responses. RaR and ARES use these scores for group relative policy optimization (GRPO), demonstrating policy training with synthesized criteria [88, 89]. Criterion-level feedback can also filter demonstrations or rank candidate responses. Dense rewards may guide learning even when final acceptance requires every critical condition to hold; the training objective and the held-out acceptance measure should therefore be reported separately.

Expert Correction of the Synthesis Process. Generated criteria can omit important conditions or reward superficial compliance, and an optimizing policy may exploit such gaps. We propose that experts inspect sampled rubrics alongside scored outputs, identify missing requirements or inappropriate weights, and feed corrections into source materials and generation instructions before regenerating affected batches. Harvey's professional-preference study illustrates lawyers critiquing model-generated criteria, but establishes an evaluation protocol rather than policy training with those criteria [90]. A repeated expert-correction loop remains a proposal here. Its value should be tested through independent assessment of unseen work, with rubric coverage and generation, scoring, and review costs reported alongside training gains.

Together, environments expand practice, synthetic reasoning expands demonstrations, and rubric synthesis expands reward specifications. Each converts limited professional input into reusable supervision. Deployment combines trained models with the information, tools, and operating arrangements needed for professional work.

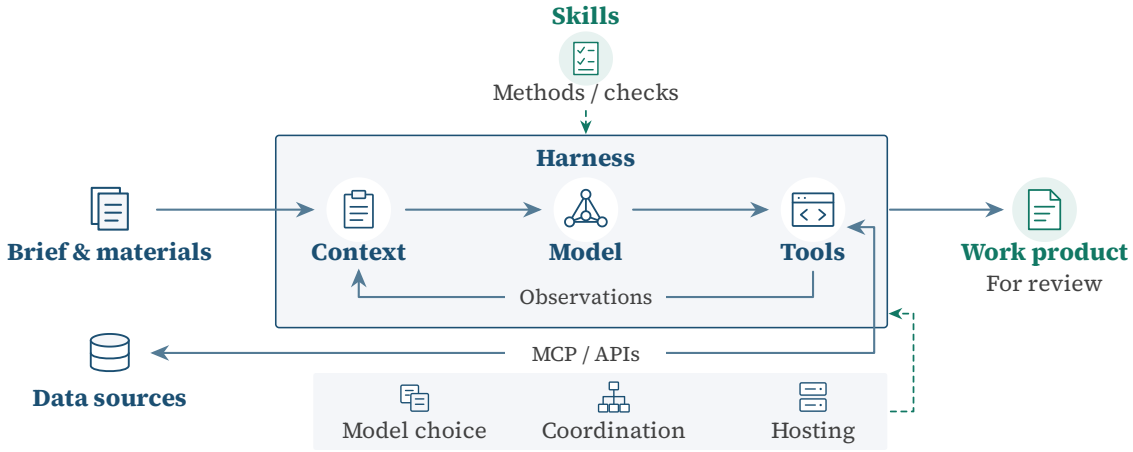


Figure 7. Deployment architecture for knowledge work. The clarified brief and case materials initialize a context that also holds retrieved evidence, assumptions, and intermediate results. Skills guide methods and checks within the model–tool execution loop. Model composition and hosting set operating conditions. The work product is submitted for professional review.

5 Deployment

Deployment connects model capabilities to the three stages of knowledge work (Section 2.2). A valuation begins with deciding what question to answer, which assumptions to examine, and what evidence is needed. Models can retrieve sources, perform calculations, and assemble a workbook, while professionals judge its suitability for the decision and remain accountable for its use. These activities require information access, reusable procedures, coordinated tool execution, and suitable model and hosting choices. Figure 7 shows the resources and execution loop supporting this work. We examine how these choices support task clarification, analysis, and professional review, with further evidence and worked cases in Appendix E.

5.1 Professional Data and Institutional Context

A request may supply a document or a topic before its objective, constraints, and evidence requirements are clear. Establishing these requirements is part of framing the case. Parameters, direct context, retrieval, and inquiry then provide complementary information as the task develops.

Knowledge Encoded in Model Parameters. Training encodes factual knowledge, such as domain terminology, concepts, and recurring relationships, in model parameters (Sections 2.1 and 4). This knowledge helps interpret obligations, relate financial quantities, and understand clinical terminology. Current case facts and authoritative versions of changing sources must be supplied at use time.

Information Supplied Directly in Context. Direct provision suits bounded material such as a contract, workbook, patient summary, or student's solution. The working context should also retain the agreed objective, constraints, and assumptions as questions are resolved. Harvey II's matter Spaces illustrate persistent project context [91]. Financial figures must remain tied to their company, reporting period, and unit, just as clauses must retain their definitions. MedAlign links timestamped clinical events with narrative notes [92].

Information Selected through Retrieval. Professional work draws on public databases, commercially licensed collections, and private institutional repositories. Public resources include company-filing collections and biomedical databases such as PubMed [49, 93]. Commercial databases supply curated financial data and legal research [94, 95]. Institutional databases and knowledge bases hold patient and client records, case documents, and internal policies [92, 96]. Combining these sources brings external evidence together with the facts and rules of the current assignment.

Structured records can be selected with SQL queries or service APIs, while document collections support keyword and semantic retrieval. Known records can be supplied directly or queried as new questions arise. FinanceBench compares full-document, answer-bearing-page, and retrieval configurations to separate evidence selection from reasoning [49]. Evidence position also affects its use [97]. Configuration choices balance coverage, processing cost, and loss of decisive details through selection or summarization.

Professional retrieval must preserve source identity and applicability. LegalBench-RAG evaluates precise passages in legal documents, and FinFIRST separately assesses financial information acquisition, source verification, and calculation [98, 99]. Structure-aware passages, exact and semantic matching, and metadata help retrieve the relevant clause or figure. Permissions, effective dates, and document versions constrain which evidence applies. In public administration, GOV.UK Chat's initial experiment encountered missing context and difficulties with long pages [100]. Retrieved passages should remain linked to the claims and case conditions they support.

Information That Remains Difficult to Obtain. Licensed research, privileged documents, and records held by another institution may lie outside the connected or authorized collection. Other information emerges through conversation. A lawyer needs the client's priorities and acceptable concessions, a clinician may need an unreported symptom, and a tutor must distinguish a misconception from a careless error. InsufficiencyBench evaluates recognition of legally material gaps in user queries [66]. Tutor CoPilot's reliance on dialogue, with whiteboard activity proposed as an additional input, illustrates how observation channels shape what the system can infer [101]. Clarifying these facts and preferences can change the questions and methods used in the current assignment.

Past work often records an outcome more fully than the reasoning behind it. A final contract may preserve the chosen clause while omitting why alternatives were rejected. Applying that precedent to a new case requires understanding the original objectives, constraints, and reasons for the choice.

5.2 Skills and Professional Procedures

A professional procedure links the question being asked to evidence gathering, analysis, and checks. In the Agent Skills specification, a Skill packages instructions in a SKILL.md file with optional scripts, references, and examples, loaded when relevant [13]. Professional methods also appear in system prompts, institutional playbooks, worked examples, and code-controlled workflows. These forms make guidance reusable at use time without changing model weights. Their usefulness depends on which decisions and conditions they preserve and how they are applied to the case.

From Expert Practice to Reusable Procedures. Written manuals and templates preserve formulas, review rules, and checks. Expert elicitation adds observations and reasons that practitioners

can articulate. BRIDGE used cognitive task analysis in which educators revised responses to student errors and explained what they noticed and why they chose a response. Its structure separates the interpretation of the error, the teaching strategy, and its intention [82]. A question intended to diagnose a misconception and one intended to prompt elaboration serve different purposes, which affects when each is useful.

SkillRefine obtains procedures from software documentation, expert edits to PIMS planning models, and construction results [102]. These records expose coordinated changes whose dependencies can be expressed as instructions and checks. Testing on new cases helps establish which relationships recur and which choices belong to one assignment. Reusable procedures need corresponding applicability conditions, required inputs, and ways to handle exceptions.

Choosing and Applying a Procedure. In statutory research, identifying the legal relationship and decisive facts determines which provisions to retrieve. The public statutory-retrieval Skill organizes factual confirmation, validity checks, application to the case, and renewed retrieval [103]. Applying it requires recognizing missing facts and revisiting the legal characterization as evidence develops. A short instruction can therefore contain a demanding judgment about what the case requires.

Applicability conditions, contrasting examples, and decision rationales make codifiable methods available for reuse (Section 2.1). In financial modeling, a DCF procedure carries assumptions through formulas and sensitivity analysis, while an analyst must justify the assumptions themselves [94]. Stable operations can follow fixed sequences. When evidence changes the case, the procedure needs questions, alternatives, and criteria for revisiting earlier choices. Modular procedures reduce irrelevant context while making module selection a consequential decision. Appendices E.3 and E.4 illustrate how financial and legal expertise change the questions, required evidence, and completion checks written into a Skill.

Effectiveness and Gaps Relative to Experts. SkillsBench v4 compares model–harness configurations with and without curated Skills. Across 87 tasks and 18 configurations, the mean pass rate increases from 33.9% to 50.5%. On finance/economics tasks, it rises from 19.1% to 33.3% [104].

BRIDGE (2024) provides an expert reference for responding to students' mathematical mistakes. GPT-4 first identifies the error, selects a teaching strategy and its intention, then writes a reply. Teachers rate alternative replies against the original tutor response on a preference scale from -2 to 2. GPT-4's mean score rises from 0.54 without the procedure to 1.02 with this self-selected decision sequence; expert-written replies score 1.26 [82]. The procedure brings GPT-4's mean feedback rating closer to the expert reference.

At the level of complete consultation workflows, AMIE (Video, 2026) combines Gemini-based dialogue, clinical planning, and perception agents. In 100 standardized scenarios enacted by patient actors, it is compared with 10 primary-care physicians, with consultations assessed by 20 physician evaluators. Its mean score on case-specific clinical rubrics is 83% of available points, versus 68% for physicians [105].

Takeaway: Professional Expertise Guides Evidence Selection and Analysis

Skills connect task framing to execution by specifying questions, evidence requirements, methods, and checks. Studies reviewed here report gains in task completion and expert-rated output quality. Reusable procedures carry professional choices into new cases, where their selection and application must follow the objectives, constraints, and available evidence.

5.3 Tool Integration and Workflow Execution

Carrying out an assignment requires operations on documents, data, and business systems. Tool calling enables individual operations, MCP connects the services that provide them, and the harness coordinates their execution within the workflow.

Tool Calling. Tool calling lets a model select an external operation and supply its arguments. The application executes the request and returns an observation that informs the next step [106]. Financial analysis can link company-filing retrieval, figure extraction, calculation, and workbook editing. The DCF workflow preserves formulas and recalculates the workbook as assumptions change [94]. Comparable operations support legal document review and clinical record processing. Intermediate results and drafts remain available for inspection and revision during the task.

Supervised tool-use trajectories and reinforcement learning develop operation selection, argument generation, and the use of returned observations [107, 108]. At deployment, tool descriptions and examples expose the available interfaces, while professional procedures and task context guide which operations are useful and how their results enter the analysis.

MCP. The Model Context Protocol (MCP) gives AI applications a common interface for discovering and invoking services' tools and accessing their resources. Providers expose their capabilities through MCP servers, and the application selects which connections and tools to make available to the model [109]. This supports reuse of integrations as professional applications combine different models, data providers, and work systems.

Professional services differ in cost and access conditions. BioMCP provides open-source tools for biomedical literature and trial data, with many workflows usable without credentials [93]. Alpha Vantage offers a free API quota and paid access for higher usage or premium market data [110]. Pkulaw uses account tokens and offers credit recharge for its statute and case services [95]. Patient-record connectors require institutional authorization [111]. Deployment costs combine data and service charges, connector hosting, and model inference.

Harness. A harness assembles context, exposes and runs tools, maintains task state, and determines when execution continues or finishes [112]. It combines fixed operations with model-directed investigation. In a valuation, the model can investigate an unexpected result while execution logic requires the workbook to recalculate before it is passed for review. Professional procedures thereby shape both the sequence of actions and the checks applied to their results.

Domain-specific design centers on the state and checks that matter to the assignment. FinanceHarness restricts evidence by publication cutoff to reconstruct the information available at a historical decision date [113]. Baichuan-M4 distinguishes the current consultation from the patient's long-term profile, retrieves underlying records as needed, and checks actions against clinical workflow constraints [114]. In each case, the harness preserves the context needed for the next professional decision and coordinates the operations that follow.

Runtime interfaces can make clarification and review explicit. LangGraph uses checkpoints to preserve state during an interrupt and resumes with external input [115]. In professional use, such pauses can obtain a missing client constraint or request review of an analysis. Operating rules should identify which choices need professional input and provide the sources, assumptions, and open questions needed to accept the result or return it for further work (Section 2.2).

The same model can perform differently across harnesses. In MedicalHarness's evaluation of five models with five harnesses on 107 tasks, harness choice and its interaction with the model

explain about a quarter of measured outcome variance [112]. Harness Scaling addresses this dependence by varying prompts, tool interfaces, observations, context management, and workflow across 18 training configurations, improving transfer to held-out configurations and external harnesses in workspace evaluations [116].

Takeaway: The Harness Organizes Execution and Review

Tool calling and MCP provide access to operations and services. The harness coordinates execution, maintains task state, and applies checks. Its operating rules should specify how missing requirements are clarified and how results reach professionals for judgment. Models should be assessed in the harness and review arrangements used for the assignment.

5.4 Specialized Models and Deployment Choices

Model and hosting choices follow the work assigned to the system, the information it can access, and the review arrangements. Deployment selects general or specialized models, distributes work among them, and locates computation. These choices affect task performance, data access, capacity, latency, and operating cost.

General and Specialized Models. GLM-Legal's October 2026 release combines legal mid-training and reinforcement learning with a professional harness [117]. On Vals' archived Harvey LAB v1 evaluation, it achieves a 19.17% task pass rate against 6.67% for its GLM-5.3-Flash base [118]. Ling-3.0-flash-Fin, released in September 2026, targets financial research and spreadsheet work. Its model-card Finance Agent v2 score falls between GPT-5.6-Sol and Gemini-3.7-Flash, while Claude-Opus-5 leads it on SpreadsheetBench v2 [119]. Specialization can improve performance over a base model while contemporary general models remain stronger on parts of the workflow.

Medical applications also distinguish reasoning from perception. MedGemma 1.5 4B supplies a compact component for medical images, laboratory reports, and health records [120]. The 2026 clinical AMIE study instead used general Gemini models with clinician-developed prompts and consultation phases to collect pre-visit histories from 100 patients [121]. The choice between parameter adaptation and application design depends on the capability needed, the available data, and the cost of maintaining the resulting system.

Combining Models. A workflow can distribute retrieval, calculation, drafting, or other subtasks among models in sequence or in parallel, with a coordinator integrating results. Handoffs need to carry sources, intermediate results, and unresolved questions. In Harvey's synthetic diligence data rooms, workers read bounded document groups and a root agent directs investigation and assembles the memorandum. In its model-pair comparison, varying the root has a larger effect than varying workers. A separate depth experiment finds that an additional delegation layer reduces average performance [122]. Routing offers another allocation mechanism, choosing a model for a request according to expected quality and cost [123]. Deployment must also specify which decisions professional reviewers retain. Compare these arrangements with a single-model workflow on delivered quality, latency, and review effort.

Cloud, Institutional, Edge, and Hybrid Deployment. Managed cloud services make provider-hosted models available on demand while the provider operates inference infrastructure. Harvey's legal platform illustrates this arrangement with regional processing and retention con-

trols [124]. Institutional deployment runs models on infrastructure controlled by the organization, such as its own servers or a private-cloud environment. It offers control over model versions and integration with restricted records, alongside responsibility for capacity, updates, and recovery. Open weights, including Ling-fin's, enable this choice [119]. Its economics depend on utilization as well as the hardware needed for weights, concurrent requests, and context caches.

Edge deployment runs models on users' devices or nearby workstations, supporting immediate or offline processing within device memory and power limits. Hybrid deployment divides work between local and remote infrastructure. MedGemma documents one such workflow, with local processing of private health information followed by anonymized requests to a central model [125]. Such a workflow must specify which facts and intermediate results cross between components and where logs and outputs are kept.

Takeaway: Match Model Choice and Deployment to the Task

Model choice, coordination, and hosting should follow the work assigned to the system and the review it receives. General and specialized models offer different strengths across tasks, while hosting shapes data access, capacity, and operational responsibility. Compare delivered quality, review effort, latency, cost, and data control.

6 Societal Impacts

AI's effects on knowledge work extend beyond productivity to how work is organized, how expertise is developed and sustained, and how benefits and costs are distributed. We focus on three dimensions: organizational structure and accountability, the reproduction of professional expertise, and impacts beyond organizational boundaries.

6.1 Organizational Structure and Accountability

Organizational structure comprises the relatively stable arrangement of roles, units, and relationships through which work is divided and authority and coordination are organized [126]. Accountability concerns who must explain and justify their conduct, to whom, and with what possible consequences [127]. AI can change who performs the work and who controls the knowledge used to perform it, without corresponding changes in who makes decisions or answers for the outcomes. This subsection examines how these changes reshape staffing, decision authority, and accountability, and how existing organizational arrangements shape AI deployment in turn.

AI Can Narrow the Staffing Pyramid. AI reduces demand for junior staff by reallocating previously delegated work to senior–AI workflows. This internal reorganization reduces the substantive work available to junior entrants [128]. The remaining work shifts from performing tasks to checking AI-generated results. Xu et al.'s model captures this shift under worker-level automation, predicting fewer but more knowledgeable juniors who verify outputs and reduce referrals to senior experts [129]. Lower demand for junior execution need not produce immediate layoffs. It can instead narrow recruitment, as observed in GenAI-adopting firms where junior employment declines through slower hiring while senior employment continues to grow [130]. Reduced hiring likewise accounts for weaker employment among young workers in AI-exposed occupations [131]. The staffing pyramid therefore narrows through changes in both work allocation and recruitment. Firms reorganize tasks around senior–AI workflows, then need fewer juniors to sustain the same workload. Section 6.2 examines the implications for training future experts.

AI Flattens Hierarchies and Redistributes Decision Authority. AI gives workers direct access to information and task support, reducing reliance on managerial consultation and coordination. AI assistance shifts software development activity from project management toward coding and from collaborative toward independent work [132]. Firms investing in their own AI capabilities tend to have smaller workforce shares of middle managers and senior employees [133]. AI also redistributes influence toward those supplying decision-relevant knowledge. Aghion and Tirole's distinction explains how effective control can shift without changing formal approval rights (Section 2.2). In a telecommunications firm, data scientists gained influence over sales-opportunity identification and helped management reallocate customers across channels, displacing account managers' influence [134]. Flattening thus reduces intermediary work and redistributes the authority previously attached to it.

AI Converts Professional Experience into Organizational Assets. AI turns parts of professional experience into resources that organizations can retain and deploy across their workforce. Expert practices become embedded in prompts, skills, and knowledge bases rather than remaining available only through their original holders. In tutoring, expert decisions are structured into guidance for AI responses and delivered to other tutors through Tutor CoPilot [82, 101]. AI-assisted customer support similarly distributes effective practices to less experienced workers [135]. These assets capture the codifiable part of professional experience; its tacit part stays with experts and surfaces only in their judgments (Section 2.1). Building these assets involves selecting which aspects of expertise to encode and how to represent them [136, 137]. Through these choices, organizations determine which expert practices enter shared workflows and how they are updated. Experience thus becomes an organizational asset not merely because it is reusable, but because the organization gains control over its distribution and application beyond the original expert's work.

Execution Moves to AI, but Accountability Remains with People and Organizations. Delegating work to AI does not transfer responsibility for outcomes. Yet control is divided among developers, deploying organizations, and professionals, allowing responsibility to be passed along the supply chain [138]. Final reviewers may then bear blame for upstream choices they cannot control, exemplifying Elish's "moral crumple zones" [139]. Human approval is meaningful only when reviewers have the evidence, time, and authority to challenge outputs and intervene. Those controlling model selection, workflow design, and deployment must also answer for these decisions, with end-to-end auditing enabling scrutiny [140]. Accountability needs to follow control, not stop at the final signature.

Organizational Structure Shapes AI Deployment. AI deployment requires cross-team coordination of technical integration, professional workflow changes, and error management. In a game-theoretic network model, managerial support and adaptable collaboration ties facilitate coordination and accelerate diffusion, while managerial resistance slows it [141]. Unassigned responsibilities can stall deployment, leaving LLMs in supporting roles when neither technical nor compliance teams accept responsibility for their risks [142]. Review procedures help manage these risks during use. McKinsey's survey associates greater reported AI value with explicit rules for when and how outputs receive human validation [143]. Turning capabilities into routine practice thus requires coordination, clear responsibilities, and review, not merely available technology.

Takeaway: AI Deployment Is an Organizational Design Choice

AI changes who performs work, whose knowledge is reused, and who influences decisions. Existing organizational arrangements shape these changes in turn. Deployment must therefore address task allocation and decision rights alongside technical capability. Retaining human approval is insufficient unless responsibility remains connected to the information, resources, and authority needed to exercise control.

6.2 Reproducing Expertise

Section 2.1 introduced the periphery problem, which arises because the codified or codifiable work through which professions train their next experts is the work AI absorbs first. This subsection defines the problem, explains why individual firms have insufficient incentives to preserve training, and distinguishes the forms of loss.

> **Definition (Periphery Problem).** Peripheral work yields two products: output, and the expertise its performers acquire by doing it under supervision. The periphery problem is the loss of the second product when AI is used to produce the first, which arises because the properties that make a task suitable for a novice also make it suitable for AI.

Training Is a Byproduct Firms Do Not Price. Entry-level work combines useful output with opportunities to learn [32, 144]. Juniors develop pattern recognition through repetition [145] and acquire shared professional knowledge by participating in cases with experienced colleagues [24]. AI separates these functions. Yu and Moon describe entry-level software tasks moving into senior–AI workflows, leaving juniors with fewer opportunities to solve problems themselves [128]. Research training has the same structure: graduate students learn to judge questions and evidence through literature reviews, data processing, and replications, tasks on which research agents are now evaluated [29]. The work still gets done, but no longer provides the same training. Replacing that lost practice requires investment whose benefits the firm may not fully recover. Skills take time to develop, and workers can carry them to other employers [146]. Hiring experienced staff avoids this investment for one firm, but cannot sustain the profession's expertise if every firm relies on others to train it [147]. Lower junior wages have traditionally helped finance learning [146], an arrangement that depends on juniors producing work worth paying for. Garicano and Rayo's model shows how automating that work can undermine the economic viability of training-based careers [148]. Once production no longer pays for learning, firms must fund training explicitly rather than assume it will follow from completing the work.

Employment Data See One Loss in Three. AI can reduce the developmental value of work in three distinct ways. In entry substitution, firms hire fewer newcomers. Labor-market evidence links weaker employment among young workers in AI-exposed occupations primarily to reduced hiring, not to separations [17, 131]. In participation substitution, newcomers remain employed but lose access to tasks that previously trained them. Junior software engineers lose entry-level tasks to senior engineers working with AI [128], while surgical residents lost opportunities to participate when robotic systems made their assistance less necessary [149]. In engagement substitution, newcomers still perform the task, but AI takes over the reasoning that supports learning. After chatbot access was removed, students who had used it without restrictions performed worse than those who had learned without AI [150]. Developers learning an unfamiliar library with AI showed weaker understanding [151]. Participation and engagement losses can

therefore occur even when hiring continues. Employment statistics capture entry losses but miss these changes within jobs. Stable employment does not establish that the learning pathway remains intact.

The Ladder Inverts. When AI takes over routine execution, newcomers may be asked to review outputs before learning to produce the work themselves. Review requires *Evaluate*-level judgment, above the *Apply*-level tasks being replaced (Section 3.1), and lies on the tacit side of the boundary in Section 2.1. Yet reliable judgment develops through practice and feedback [152]. Learning to assess a draft, for example, involves producing drafts and understanding how experienced colleagues correct them. Giving newcomers approval rights before they acquire this experience separates formal authority from the knowledge needed to exercise it, contributing to authority drift (Section 2.2). The problem grows as expertise passes between generations. Today's seniors can draw on experience gained by doing the work now assigned to AI, while their successors may lack that foundation. This also weakens the expert signal used in evaluation and training (Section 5.1). Approval becomes less informative when reviewers have had fewer opportunities to develop independent judgment.

The Cost Is Uneven and Delayed. The loss depends on how expertise is learned and how quickly mistakes become visible. Codified knowledge and rapid feedback provide alternative learning routes. In software, tests and public code offer opportunities to practise outside assigned work. Alternatives are harder to provide when expertise depends on perceptual and collective knowledge and outcomes take longer to assess (Section 2.1). Formal pathways such as residency and articling preserve structured training, but a training position alone does not ensure meaningful participation, as the surgical case illustrates [149]. The cost also takes time to emerge. Current experts can sustain performance while the pathways producing their successors weaken [153, 154]. Training must therefore be funded and organized explicitly once production no longer provides it as a byproduct. Juniors can investigate anomalous AI outputs and explain their judgments to seniors [155]. Training costs can be shared across firms [156], with industry bodies coordinating pooled funding and joint training [157]. Progress should be measured by meaningful practice, not simply by the number of juniors employed.

Takeaway: Completing the Work Does Not Train the Next Expert

AI can preserve production while removing the practice through which newcomers develop judgment. Stable hiring does not reveal whether that practice survives. Once learning ceases to be a byproduct of productive work, it must be deliberately organized and funded. Training should be assessed through meaningful participation and independent judgment, not headcount alone.

6.3 Impacts Beyond Organizational Boundaries

Decisions to deploy AI are made within organizations, but their consequences extend to people and shared resources beyond their boundaries. Organizations do not bear all the costs their choices create. This subsection examines these external effects and argues that organizational gains cannot be assessed independently of the burdens placed on others.

Direct AI Advice Does Not Carry the Safeguards of Professional Service. Consulting a standalone chatbot does not by itself establish a professional–client relationship. Licensing imposes competence standards and disciplinary oversight, while fiduciary duties bind advisers to clients'

interests [158]. Professional-looking AI answers invite trust without automatically carrying these obligations [159]. The potential to widen access to advice does not close this protection gap [158, 160]. In U.S. legal practice, chatbot conversations do not automatically receive attorney–client privilege, and privacy promises do not provide equivalent protection against compelled disclosure [161]. Direct AI services need enforceable safeguards for competence, loyalty, and sensitive disclosures. Better answers and broader access cannot substitute for these protections.

AI Shifts the Costs of Professional Restructuring onto Practitioners. When AI replaces professional tasks, organizations save on execution while practitioners bear income losses and career-transition costs. Among freelancers in AI-exposed occupations, employment and earnings decline, with no evidence that prior performance protects against employment losses [162]. Moving to another occupation also puts accumulated expertise at risk. Experience earns occupation-specific returns [163], so finding a new job does not necessarily restore the value of previous professional investment. Nor does moving to another occupation reliably protect workers from declining demand. Workers leaving declining, AI-exposed occupations disproportionately enter other declining rather than growing occupations [164]. Reemployment therefore does not establish recovery. Lost earnings, retraining time, and diminished returns on expertise remain costs to practitioners even when organizations report productivity gains.

AI Concentrates Returns among Model and Platform Providers. Providers repeatedly sell AI services built from professional expertise, spreading development costs across customers and gaining economies of scale [165]. Professional contributors, by contrast, may be paid for their input without sharing subsequent revenue. OpenAI's Project Mercury reportedly pays former investment bankers about $150 per hour to produce financial analyses for AI training [166]. Such fees reward their work but do not themselves grant rights to continuing returns. Some creative workers' demands for additional payment if their work is reused after employment ends reflect this gap [167]. Reuse also creates competition when AI services substitute for contributors' work. Senftleben distinguishes compensation for this substitution from payment for training inputs [168]. Compensation must therefore address continuing commercial use, not treat payment for the initial contribution as settling every subsequent claim.

AI Erodes the Knowledge Commons It Learns From. AI redirects knowledge-seeking from public communities into private conversations. Public questions produce answers that others can inspect, correct, and reuse. Private exchanges do not automatically replenish that resource. Declining Stack Overflow activity after ChatGPT's release documents the weakening of public participation [169, 170]. Fewer exchanges mean fewer opportunities to record unfamiliar problems, correct mistakes, and recognize contributors. The loss concerns the renewal of shared knowledge, not simply website traffic. Acemoglu et al.'s model explains how AI-induced reductions in individual learning weaken this renewal [171]. As professional practices change, existing archives cannot replace continuing investigation and correction. Providers must support the communities producing fresh knowledge and not treat their accumulated contributions as a training resource that maintains itself.

Takeaway: Organizational Gains Do Not Settle the Social Costs

Organizations can benefit from AI while others bear losses in protection, professional income, and control over contributed knowledge. These gains also depend on shared knowledge that contin-

ued use does not automatically replenish. Assessing deployment therefore requires accounting for costs beyond the adopting organization and sustaining the contributions on which future capability depends.

7 Outlook and Conclusion

This survey has reviewed research on AI in knowledge work stage by stage. LLMs increasingly handle codified knowledge work, and codifiable work once its success criteria are written down; the main difficulty lies in tacit knowledge work, whose success criteria those who apply them cannot state in advance, and within it at the two ends of a task. Framing and planning, chiefly *Create*, has received the least attention; execution, chiefly *Understand*, *Apply*, and *Analyze*, is where research has concentrated; and judgment and accountability, chiefly *Evaluate*, is mostly simulated with rubrics written in advance. We therefore propose three directions for research.

Evaluate Framing and Judgment. Tasks at the *Create* and *Evaluate* levels are the least covered by current evaluations. Such tasks often have no single correct answer. Evaluations can borrow the way the professions assess newcomers, judging how much supervision a model's work still needs [172], and should report disagreement among experts as it is.

Learn from Expert Practice. Research on knowledge in LLMs has extended from facts to procedures; the next step is expert experience. When experts compare, revise, or return work, tacit knowledge surfaces in partial form. For agents to learn from these signals, corrections must be separated from preferences, and credit must be assigned when outcomes arrive late; with these in place, deployment and training can form a learning loop anchored in expert judgment.

Collaborate and Sustain Expertise. Execution, whose results can be checked, is well suited to agents. Framing and planning, and judgment and accountability, are still led by professionals, and how humans and agents should collaborate in these stages remains open: for example, agents could propose possible directions and options, experts could choose among them, and those choices could serve as learning signals. Whatever the division of labor, real authority needs to stay with those who bear responsibility, and the same act of supervision can serve both to train newcomers and to train agents.

Taken together, our analysis supports a division of labor between agents and professionals. Execution, whose results can be checked, will be increasingly automated; framing and judgment should become collaborative, with agents proposing and professionals deciding and answering for the result. This division is also a condition of further progress: expert choices in collaboration are the signals from which agents learn tacit knowledge, and they persist only while professionals keep exercising judgment and newcomers keep acquiring it. Realizing it requires evaluations that cover all three stages, training that learns from expert choices, and organizations that align authority with accountability and preserve paths for training newcomers. We hope this framework helps researchers understand where agents stand in knowledge work, and helps professionals and organizations think about how to work alongside them.

References

[1] Michael Polanyi. *The Tacit Dimension*. Doubleday, Garden City, NY, 1966.

[2] Daniel Martin Katz, Michael James Bommarito, Shang Gao, and Pablo Arredondo. GPT-4 Passes the Bar Exam. *Philosophical Transactions of the Royal Society A*, 382(2270):20230254, 2024. doi: 10.1098/rsta.2023.0254.

[3] Karan Singhal et al. Large Language Models Encode Clinical Knowledge. *Nature*, 620 (7972):172–180, 2023. doi: 10.1038/s41586-023-06291-2.

[4] Tejal Patwardhan et al. GDPval: Evaluating AI Model Performance on Real-World Economically Valuable Tasks, 2025. URL https://arxiv.org/html/2510.04374.

[5] Bertie Vidgen et al. APEX-Agents, 2026. URL https://arxiv.org/html/2601.14242v3. Version 3, revised February 23, 2026; performance comparisons refer to the models and configuration tested in this version.

[6] Peter F. Drucker. *Landmarks of Tomorrow*. Harper & Brothers, New York, 1959.

[7] David Autor. Polanyi's Paradox and the Shape of Employment Growth. NBER Working Paper 20485, National Bureau of Economic Research, 2014. URL https://www.nber.org/papers/w20485.

[8] Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. Language Models as Knowledge Bases? In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2463–2473. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1250.

[9] Adam Roberts, Colin Raffel, and Noam Shazeer. How Much Knowledge Can You Pack Into the Parameters of a Language Model? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5418–5426. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.emnlp-main.437. URL https://aclanthology.org/2020.emnlp-main.437/.

[10] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks, 2020. URL https://arxiv.org/abs/2005.11401.

[11] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An Open-Ended Embodied Agent with Large Language Models, 2023. URL https://arxiv.org/abs/2305.16291.

[12] Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. Agent Workflow Memory, 2024. URL https://arxiv.org/abs/2409.07429.

[13] Agent Skills. Agent Skills specification. Open specification, 2026. URL https://agentskills.io/specification. Accessed October 23, 2026.

[14] John Yang, Kilian Lieret, Carlos E. Jimenez, Alexander Wettig, Kabir Khandpur, Yanzhe Zhang, Binyuan Hui, Ofir Press, Ludwig Schmidt, and Diyi Yang. SWE-smith: Scaling data for software engineering agents, 2025. URL https://arxiv.org/abs/2504.21798v1. Cited version: arXiv:2504.21798v1.

[15] Bengt Holmström and Paul Milgrom. Multitask Principal–Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design. *Journal of Law, Economics, & Organization*, 7:24–52, 1991.

[16] Michael Polanyi. *Personal Knowledge: Towards a Post-Critical Philosophy*. University of Chicago Press, Chicago, 1958.

[17] Maxim Massenkoff and Peter McCrory. Labor Market Impacts of AI: A New Measure and Early Evidence. Research report, Anthropic, March 2026. URL https://www.anthropic.com/research/labor-market-impacts.

[18] Peter F. Drucker. Knowledge-Worker Productivity: The Biggest Challenge. *California Management Review*, 41(2):79–94, 1999. doi: 10.2307/41165987.

[19] David H. Autor, Frank Levy, and Richard J. Murnane. The skill content of recent technological change: An empirical exploration. *The Quarterly Journal of Economics*, 118(4):1279–1333, 2003. doi: 10.1162/003355303322552801.

[20] Badr AlKhamissi, Millicent Li, Asli Celikyilmaz, Mona Diab, and Marjan Ghazvininejad. A Review on Language Models as Knowledge Bases, 2022. URL https://arxiv.org/abs/2204.06031.

[21] Gilbert Ryle. *The Concept of Mind*. Hutchinson, London, 1949.

[22] Lorin W. Anderson and David R. Krathwohl, editors. *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*. Longman, New York, 2001.

[23] Robin Cowan, Paul A. David, and Dominique Foray. The explicit economics of knowledge codification and tacitness. *Industrial and Corporate Change*, 9(2):211–253, 2000. doi: 10.1093/icc/9.2.211.

[24] Harry Collins. *Tacit and Explicit Knowledge*. University of Chicago Press, Chicago, 2010. doi: 10.7208/chicago/9780226113821.001.0001.

[25] Haridimos Tsoukas. Do We Really Understand Tacit Knowledge? In *Complex Knowledge: Studies in Organizational Epistemology*, pages 141–161. Oxford University Press, Oxford, 2005. doi: 10.1093/oso/9780199275571.003.0007.

[26] Vals AI. Vals AI Enterprise Benchmarks: Corporate Finance, Mortgage Tax, Case Law, and Legal Research. Online documentation, 2026. URL https://www.vals.ai/benchmarks. Accessed 2026-10-14.

[27] Harvey. Introducing Harvey's Legal Agent Benchmark. Online documentation, 2026. URL https://www.harvey.ai/blog/introducing-harveys-legal-agent-benchmark. Accessed 2026-10-14.

[28] David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A Graduate-Level Google-Proof Q&A Benchmark, 2023. URL https://arxiv.org/abs/2311.12022.

[29] Giulio Starace et al. PaperBench: Evaluating AI's Ability to Replicate AI Research, 2025. URL https://arxiv.org/html/2504.01848.

[30] Takashi Onoda and Yasunobu Ito. What Aspects of Tacit Knowledge Are Structurally Excluded from Generative AI? A Conceptual Framework of Mediation, Structure, and Representation. In *The Human Side of Service Engineering*, volume 221 of *AHFE International*. AHFE International, 2026. doi: 10.54941/ahfe1007726.

[31] Mina Cho, Russell J. Funk, Alok Gupta, and Mochen Yang. Large Language Models Can Explain Experts Better Than Experts Themselves, 2026. URL https://arxiv.org/abs/2608.07488v2. Version 2, revised October 21, 2026.

[32] Jean Lave and Etienne Wenger. *Situated Learning: Legitimate Peripheral Participation*. Cambridge University Press, Cambridge, 1991. doi: 10.1017/CBO9780511815355.

[33] Donald A. Schön. *The Reflective Practitioner: How Professionals Think in Action*. Basic Books, New York, 1983.

[34] Eugene F. Fama and Michael C. Jensen. Separation of Ownership and Control. *The Journal of Law and Economics*, 26(2):301–325, 1983.

[35] Philippe Aghion and Jean Tirole. Formal and Real Authority in Organizations. *Journal of Political Economy*, 105(1):1–29, 1997.

[36] Yining Hua, Hongbin Na, Cyrus Ayubcha, and Levi Lian. Designing Benchmarks for Knowledge Work, 2026. URL https://arxiv.org/html/2605.23262v2.

[37] Rahul K. Arora et al. HealthBench: Evaluating Large Language Models Towards Improved Human Health, 2025. URL https://arxiv.org/abs/2505.08775.

[38] Yixing Jiang et al. MedAgentBench: A Realistic Virtual EHR Environment to Benchmark Medical LLM Agents, 2025. URL https://arxiv.org/abs/2501.14654.

[39] Qianchu Liu et al. HealthAgentBench: A Unified Benchmark Suite of Realistic Agentic Healthcare Environments for Challenging Frontier AI Agents, 2026. URL https://arxiv.org/html/2606.31179.

[40] Kung-Hsiang Huang et al. CRMArena-Pro: Holistic Assessment of LLM Agents Across Diverse Business Scenarios and Interactions, 2025. URL https://arxiv.org/html/2505.18878v1.

[41] Yuning Wu et al. WritingBench: A Comprehensive Benchmark for Generative Writing, 2025. URL https://arxiv.org/html/2503.05244.

[42] Mantas Mazeika et al. Remote Labor Index: Measuring AI Automation of Remote Work, 2025. URL https://arxiv.org/html/2510.26787.

[43] Yuetai Li et al. JobBench: Aligning Agent Work with Human Will, 2026. URL https://arxiv.org/html/2605.26329v1.

[44] Bo Zhao et al. UniPPTBench: A Unified Benchmark for Presentation Generation Across Diverse Input Settings, 2026. URL https://arxiv.org/html/2605.17356v1.

[45] William G. Ouchi. A Conceptual Framework for the Design of Organizational Control Mechanisms. *Management Science*, 25(9):833–848, 1979.

[46] Jing Wu et al. OS-Marathon: Benchmarking Computer-Use Agents on Vast-Horizon, Repetitive Tasks, 2026. URL https://arxiv.org/html/2601.20650.

[47] Liqiang Jing et al. DSBench: How Far Are Data Science Agents from Becoming Data Science Experts?, 2024. URL https://github.com/LiqiangJing/DSBench.

[48] Yiyou Sun et al. Agents' Last Exam, 2026. URL https://arxiv.org/abs/2606.05405v2.

[49] Pranab Islam et al. FinanceBench: A New Benchmark for Financial Question Answering, 2023. URL https://arxiv.org/abs/2311.11944.

[50] Haoyu Dong et al. Finch: Benchmarking Finance & Accounting across Spreadsheet-Centric Enterprise Workflows, 2025. URL https://arxiv.org/html/2512.13168.

[51] Frank F. Xu et al. TheAgentCompany: Benchmarking LLM Agents on Consequential Real World Tasks, 2024. URL https://arxiv.org/html/2412.14161v2.

[52] Zhiyu Chen et al. FinQA: A Dataset of Numerical Reasoning over Financial Data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–3711. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.emnlp-main.300. URL https://aclanthology.org/2021.emnlp-main.300/.

[53] Olly Styles and Sam Miller. WorkBench Revisited: Workplace Agents Two Years On, 2026. URL https://arxiv.org/html/2606.13715v2.

[54] Artificial Analysis. Intelligence Benchmarking: GDPval-AA Methodology. Online documentation, 2026. URL https://artificialanalysis.ai/methodology/intelligence-benchmarking#gdpval-aa. Accessed 2026-10-14.

[55] Harvey. Harvey LAB: Evaluation Strategies and Scoring Documentation. Online documentation, 2026. URL https://github.com/harveyai/harvey-labs/blob/main/docs/eval-strategies.md. Accessed 2026-10-14.

[56] Bertie Vidgen et al. The AI Productivity Index (APEX), 2025. URL https://arxiv.org/html/2509.25721.

[57] Anthropic. The Anthropic Economic Index. Public dataset and technical report, 2026. URL https://huggingface.co/datasets/Anthropic/EconomicIndex.

[58] Economic Innovation Group. AI Adoption in the Census Business Trends and Outlook Survey (BTOS). Public replication repository, 2025. URL https://github.com/EIG-Research/ai-btos.

[59] Alex Wang et al. BigFinanceBench: A Workflow-Grounded Benchmark for Financial-Research Agents, 2026. URL https://arxiv.org/html/2606.03829.

[60] Shreya Shankar et al. Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences, 2024. URL https://arxiv.org/html/2404.12272.

[61] Thomson Yen et al. MBABench: Evaluating LLM Agents on End-to-End Spreadsheet Tasks in Finance, 2026. URL https://arxiv.org/html/2605.22664.

[62] Jiayuxuan Yang, Jie M. Zhang, Yiling Lou, and Zhenpeng Chen. Mubric: Mutation Testing-Guided Rubric Generation for LLM Evaluation, 2026. URL https://arxiv.org/html/2609.37322v1.

[63] Jianlin Chen, Wenhui Chen, Ziyao Lin, and Chi Man Vong. A Judge Should Know What Changed: Construct Validity for LLM-as-a-Judge Evaluation, 2026. URL https://arxiv.org/html/2608.24419v1.

[64] Yijia Shao, Vinay Samuel, Yucheng Jiang, John Yang, and Diyi Yang. Collaborative Gym: A Framework for Enabling and Evaluating Human-Agent Collaboration, 2024. URL https://arxiv.org/html/2412.15701v6.

[65] Yijia Shao et al. CollabSkill: Evaluating Human-Agent Collaboration On Real-World Tasks, 2026. URL https://arxiv.org/html/2606.09833.

[66] Samuel J. Vincent, Daniel Calloway, Fangyi Yu, Andrew M. Bean, and Nabeel Seedat. InsufficiencyBench: Evaluating LLM Legal Advice on Underspecified User Queries, 2026. URL https://arxiv.org/abs/2608.20220.

[67] Christine Herlihy, Jennifer Neville, Tobias Schnabel, and Adith Swaminathan. On Overcoming Miscalibrated Conversational Priors in LLM-based ChatBots. In *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, volume 244 of *Proceedings of Machine Learning Research*, pages 1599–1620. PMLR, 2024. URL https://proceedings.mlr.press/v244/herlihy24a.html.

[68] Hanglong Lv, Dawei Zhu, Lei Li, Bowen Ye, Huaqiu Liu, Yifan Song, Bofei Gao, Weimin Xiong, Jinhao Dong, Chenhong He, Lingpeng Kong, Qi Liu, Tong Yang, and Fuli Luo. PersonaForge: Realistic Multi-Turn User Simulation for Agentic Systems. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, pages 46647–46670. Association for Computational Linguistics, 2026. URL https://aclanthology.org/2026.findings-emnlp.2181/.

[69] Haohao Luo, Zexi Li, Yuexiang Xie, Wenhao Zhang, Yaliang Li, and Ying Shen. IntentRL: Training Proactive User-intent Agents for Open-ended Deep Research via Reinforcement Learning. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 82557–82584. PMLR, 2026. URL https://proceedings.mlr.press/v306/luo26p.html.

[70] Bowen Ye, Lei Li, Shicheng Li, Zihao Yue, Linghao Zhang, Hanglong Lv, Yuanxin Liu, Wenhan Ma, Hao Tian, Rang Li, Jinhao Dong, Yikai Zhao, Xiangwei Deng, Hailin Zhang, Liang Zhao, Qi Liu, Lingpeng Kong, Tong Yang, and Fuli Luo. CodeMidas: Scaling agentic coding RL environments from code itself, 2026. URL https://arxiv.org/abs/2609.22068v1. Cited version: arXiv:2609.22068v1.

[71] Dunwei Tu, Hongyan Hao, Hansi Yang, Yihao Chen, Yi-Kai Zhang, Zhikang Xia, Yu Yang, Yueqing Sun, Xingchen Liu, Furao Shen, Qi Gu, Hui Su, and Xunliang Cai. ScaleEnv: Scaling environment synthesis from scratch for generalist interactive tool-use agent training, 2026. URL https://arxiv.org/abs/2602.06820v1. Cited version: arXiv:2602.06820v1.

[72] Shihao Cai, Runnan Fang, Jialong Wu, Baixuan Li, Xinyu Wang, Yong Jiang, Liangcai Su, Liwen Zhang, Wenbiao Yin, Zhen Zhang, Fuli Feng, Pengjun Xie, and Xiaobin Wang. AutoForge: Automated environment synthesis for agentic reinforcement learning, 2025. URL https://arxiv.org/abs/2512.22857v1. Cited version: arXiv:2512.22857v1.

[73] Zhaorun Chen, Zhuokai Zhao, Kai Zhang, Bo Liu, Qi Qi, Yifan Wu, Tarun Kalluri, Sara Cao, Yuanhao Xiong, Haibo Tong, Huaxiu Yao, Hengduo Li, Jiacheng Zhu, Xian Li, Dawn Song, Bo Li, Jason Weston, and Dat Huynh. Scaling agent learning via experience synthesis, 2025. URL https://arxiv.org/abs/2511.03773v1. Cited version: arXiv:2511.03773v1.

[74] Ankush Agarwal, Harsh Vishwakarma, Suraj Nagaje, and Chaitanya Devaguptapu. EnterpriseLab: A full-stack platform for developing and deploying agents in enterprises, 2026. URL https://arxiv.org/abs/2603.21630v1. Cited version: arXiv:2603.21630v1.

[75] Sushant Mehta, Logan Ritchie, Suhaas Garre, Ian Niebres, Nick Heiner, and Edwin Chen. EnterpriseBench Corecraft: Training Generalizable Agents on High-Fidelity RL Environments, 2026. URL https://arxiv.org/abs/2602.16179v5. Cited version: arXiv:2602.16179v5.

[76] Suhana Bedi, Ryan Welch, Ethan Steinberg, Michael Wornow, Taeil Matthew Kim, Haroun Ahmed, Peter Sterling, Bravim Purohit, Qurat Akram, Angelic Acosta, Esther Nubla, Pritika Sharma, Michael A. Pfeffer, Sanmi Koyejo, and Nigam H. Shah. HealthAdminBench: Evaluating Computer-Use Agents on Healthcare Administration Tasks, 2026. URL https://arxiv.org/abs/2604.09937v1. Cited version: arXiv:2604.09937v1.

[77] Banghao Chi, Yining Xie, Mingyuan Wu, Jingcheng Yang, Jize Jiang, Zhaoheng Li, Shengyi Qian, Minjia Zhang, Klara Nahrstedt, Rui Hou, Xiangjun Fan, and Hanchao Yu. Spreadsheet-RL: Advancing Large Language Model Agents on Realistic Spreadsheet Tasks via Reinforcement Learning, 2026. URL https://arxiv.org/abs/2605.22642v1. Cited version: arXiv:2605.22642v1.

[78] Elaine Lau, Markus Dücker, Ronak Chaudhary, Hui Wen Goh, Rosemary Wei, Vaibhav Kumar, Saed Qunbar, Guram Gogia, Yi Liu, Scott Millslagle, Nasim Borazjanizadeh, Ulyana Tkachenko, Samuel Eshun Danquah, Collin Schweiker, Vijay Karumathil, Asrith Devalaraju, Varsha Sandadi, Haemi Nam, Punit Arani, Ray Epps, Abdullah Arif, Sahil Bhaiwala, Curtis Northcutt, Skyler Wang, Anish Athalye, Jonas Mueller, and Francisco Guzmán. BankerToolBench: Evaluating AI Agents in End-to-End Investment Banking Workflows, 2026. URL https://arxiv.org/abs/2604.11304v1. Cited version: arXiv:2604.11304v1.

[79] Lingfei Qian, Weipeng Zhou, Yan Wang, Xueqing Peng, Han Yi, Yilun Zhao, Jimin Huang, Qianqian Xie, and Jian yun Nie. Fino1: On the Transferability of Reasoning-Enhanced LLMs and Reinforcement Learning to Finance, 2025. URL https://arxiv.org/abs/2502.08127v3. Cited version: arXiv:2502.08127v3, revised 14 June 2025.

[80] Haitao Li, Yifan Chen, Shuo Miao, Qian Dong, Jia Chen, Yiran Hu, Junjie Chen, Minghao Qin, Yueyue Wu, Yujia Zhou, Qingyao Ai, Yiqun Liu, Cheng Luo, Quan Zhou, Ya Zhang, and Jikun Hu. LegalOne: A Family of Foundation Models for Reliable Legal Reasoning, 2026. URL https://arxiv.org/abs/2602.00642v2. Cited version: arXiv:2602.00642v2, revised 3 February 2026.

[81] Juncheng Wu, Wenlong Deng, Xingxuan Li, Sheng Liu, Taomian Mi, Yifan Peng, Ziyang Xu, Yi Liu, Hyunjin Cho, Chang-In Choi, Yihan Cao, Hui Ren, Xiang Li, Xiaoxiao Li, and Yuyin Zhou. MedReason: Eliciting Factual Medical Reasoning Steps in LLMs via Knowledge Graphs, 2025. URL https://arxiv.org/abs/2504.00993v1. Cited version: arXiv:2504.00993v1, submitted 1 April 2025.

[82] Rose E. Wang, Qingyang Zhang, Carly Robinson, Susanna Loeb, and Dorottya Demszky. Bridging the novice-expert gap via models of decision-making: A case study on remediating math mistakes. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2174–2199, 2024. doi: 10.18653/v1/2024.naacl-long.120. URL https://aclanthology.org/2024.naacl-long.120/.

[83] Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. STaR: Bootstrapping Reasoning With Reasoning, 2022. URL https://arxiv.org/abs/2203.14465v2. Cited version: arXiv:2203.14465v2.

[84] Xiaoke Zhao, Zhaowen Zhou, Lin Chen, Lihong Wang, Zhiyi Huang, Kaiyuan Zheng, Yanjun Zheng, Xiyang Du, Longfei Liao, Jiawei Liu, Xiang Qi, Bo Zhang, Peng Zhang, Zhe Li, and Wei Wang. Agentar-DeepFinance-300K: A Large-Scale Financial Dataset via Systematic Chain-of-Thought Synthesis Optimization, 2025. URL https://arxiv.org/abs/2507.12901v1. Cited version: arXiv:2507.12901v1, submitted 17 July 2025.

[85] Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's Verify Step by Step, 2023. URL https://arxiv.org/abs/2305.20050.

[86] Jérémy Scheurer, Jon Ander Campos, Tomasz Korbak, Jun Shern Chan, Angelica Chen, Kyunghyun Cho, and Ethan Perez. Training language models with language feedback at scale, 2023. URL https://arxiv.org/abs/2303.16755.

[87] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting, 2023. URL https://arxiv.org/abs/2305.04388v2. Cited version: arXiv:2305.04388v2.

[88] Anisha Gunjal, Anthony Wang, Elaine Lau, Vaskar Nath, Yunzhong He, Bing Liu, and Sean Hendryx. Rubrics as Rewards: Reinforcement Learning Beyond Verifiable Domains, 2025. URL https://arxiv.org/abs/2507.17746v2. Cited version: arXiv:2507.17746v2, revised 3 October 2025.

[89] Xiaoyuan Li, Keqin Bao, Moxin Li, Yubo Ma, Yichang Zhang, Wenjie Wang, Fuli Feng, and Dayiheng Liu. ARES: Automated Rubric Synthesis for Scalable LLM Reinforcement Learning, 2026. URL https://arxiv.org/abs/2605.23454v2. Cited version: arXiv:2605.23454v2, revised 25 May 2026.

[90] Julio Pereyra. Augmenting human preference in complex domains. Harvey Research blog, September 2026. URL https://www.harvey.ai/blog/augmenting-human-preference-in-complex-domains. Technical blog, September 17, 2026. Accessed October 19, 2026.

[91] Harvey Team. Introducing Harvey II. Harvey product documentation, August 2026. URL https://www.harvey.ai/blog/introducing-harvey-ii. August 18, 2026. Accessed October 23, 2026.

[92] Scott L. Fleming, Alejandro Lozano, William J. Haberkorn, et al. MedAlign: A clinician-generated dataset for instruction following with electronic medical records, 2023. URL https://arxiv.org/abs/2308.14089v2. Cited version: v2, December 24, 2023.

[93] BioMCP Contributors. BioMCP: Biomedical data access for agents. Source code and documentation, 2026. URL https://github.com/genomoncology/biomcp/tree/4050fdc7dbdf4d713d94cf02a39a147b863665be. MIT license; API-key and source-licensing documentation checked. Fixed commit 4050fdc7dbdf4d713d94cf02a39a147b863665be; accessed October 25, 2026.

[94] Anthropic. Claude for financial services: Reference agents, skills, and data connectors. Source code and documentation, 2026. URL https://github.com/anthropics/financial-services/tree/574ed3624aebd0418c7e96cd101262f30210ab26. Commit 574ed3624aebd0418c7e96cd101262f30210ab26; repository README and DCF Model Builder Skill. Accessed October 23, 2026.

[95] Pkulaw. Pkulaw MCP legal services platform. Provider documentation, 2026. URL https://mcp.pkulaw.com/. Public integration examples, token access, and credit-recharge entry. Accessed October 25, 2026.

[96] Microsoft. Microsoft Copilot overview. Microsoft Learn technical documentation, 2026. URL https://learn.microsoft.com/en-us/microsoft-365/copilot/copilot-overview. Updated October 23, 2026. Accessed October 24, 2026.

[97] Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638. URL https://aclanthology.org/2024.tacl-1.9/.

[98] Nicholas Pipitone and Ghita Houir Alami. LegalBench-RAG: A benchmark for retrieval-augmented generation in the legal domain, 2024. URL https://arxiv.org/abs/2408.10343v1.

[99] Ant Group. FinFIRST: Financial information retrieval, sourcing and traceability, September 2026. URL https://huggingface.co/datasets/inclusionAI/FinFIRST/blob/1d062a63d6e58b398d162658ffd49731f12512a6/README.md. Official dataset card; 123 tasks developed with professional support from CICC; reviewed 25 October 2026.

[100] Government Digital Service. The findings of our first generative AI experiment: GOV.UK Chat. Inside GOV.UK, 2024. URL https://insidegovuk.blog.gov.uk/2024/01/18/the-findings-of-our-first-generative-ai-experiment-gov-uk-chat/. January 18, 2024. Accessed October 24, 2026.

[101] Rose E. Wang, Ana T. Ribeiro, Carly D. Robinson, Susanna Loeb, and Dora Demszky. Tutor CoPilot: A human-AI approach for scaling real-time expertise. arXiv preprint arXiv:2410.03017, 2024. URL https://arxiv.org/abs/2410.03017v3. Version 3, revised October 10, 2026; first submitted October 3, 2024.

[102] Dongxiao Liu, Boren Zhu, Yuwen Ding, Linghui Li, Lei Li, and Xiaoyong Li. SkillRefine: Cross-source skill induction and execution validation for LLM agents in refinery planning software, 2026. URL https://arxiv.org/abs/2609.30674v1. Cited version: v1, September 25, 2026.

[103] Legal-Skills-Chinese Contributors. Legal-Skills-Chinese: Chinese legal reasoning skills. Source code and documentation, 2026. URL https://github.com/THUYRan/Legal-Skills-Chinese/tree/c31ef1f022be6bcb3e31c1a2374d1d8240ffa61f. README and statutory-retrieval Skill; fixed commit c31ef1f022be6bcb3e31c1a2374d1d8240ffa61f. Accessed October 24, 2026.

[104] Xiangyi Li, Yimin Liu, Wenbo Chen, et al. SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks, 2026. URL https://arxiv.org/html/2602.12670v4.

[105] Mahvish Nagda, Jihyeon Lee, Matthew Thompson, et al. Towards expert-level medical AI for real-time video consultations, 2026. URL https://arxiv.org/abs/2608.09861v1. Cited version: v1, 10 August 2026.

[106] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.03629.

[107] Yujia Qin, Shihao Liang, Yining Ye, et al. ToolLLM: Facilitating large language models to master 16000+ real-world APIs, 2023. URL https://arxiv.org/abs/2307.16789.

[108] Cheng Qian, Emre Can Acikgoz, Qi He, et al. ToolRL: Reward is all tool learning needs, 2025. URL https://arxiv.org/abs/2504.13958v1.

[109] Model Context Protocol Contributors. Model Context Protocol: Architecture overview. Protocol documentation, 2026. URL https://modelcontextprotocol.io/docs/2026-07-28/learn/architecture. Protocol documentation version 2026-07-28. Accessed October 23, 2026.

[110] Alpha Vantage. Customer support: API access, usage limits, and market data. Provider documentation, 2026. URL https://www.alphavantage.co/support/. Access and pricing conditions observed in the support FAQ; subject to change. Accessed October 25, 2026.

[111] Anthropic. Claude for healthcare: Connectors and workflow reference implementations. Source code and documentation, 2026. URL https://github.com/anthropics/healthcare/tree/c7ed150d376bdd13219eb0bfeb4203e12da85516. MCP configuration, experimental FHIR connector, and clinical-note extraction workflow. Fixed commit c7ed150d376bdd13219eb0bfeb4203e12da85516; accessed October 25, 2026.

[112] Ziqing Wang, Lili Zhao, and Kaize Ding. MedicalHarness: A controlled evaluation of LLMs and agent harnesses on medical tasks, 2026. URL https://arxiv.org/abs/2610.05778v1.

[113] Yijia Xiao, Rujun Han, Yanfei Chen, et al. FinanceHarness: Autonomous financial deep research framework, 2026. URL https://arxiv.org/abs/2607.27853v1.

[114] Aiyuan Yang, Canbin Piao, Chengfeng Dou, et al. Baichuan-M4: A clinical-grade medical agent system for continuous care, 2026. URL https://arxiv.org/abs/2606.08982v2. Cited version: v2, 9 June 2026.

[115] LangChain. LangGraph: Interrupts. Technical documentation, 2026. URL https://docs.langchain.com/oss/python/langgraph/interrupts. State persistence, external input, and resuming execution for human-in-the-loop workflows. Accessed October 31, 2026.

[116] Fei Bai, Huatong Song, Chuan Hao, Ran Tao, Jian Yang, Bryan Dai, Wayne Xin Zhao, and Ji-Rong Wen. Harness scaling for generalizable agentic reinforcement learning, 2026. URL https://arxiv.org/abs/2610.29458v1. Cited version: v1, October 21, 2026.

[117] Z.ai. GLM-Legal: Advancing agentic legal intelligence, October 2026. URL https://z.ai/blog/glm-legal. Official release and training description, 12 October 2026; accessed 25 October 2026.

[118] Vals AI. Harvey's legal agent benchmark v1 leaderboard, 2026. URL https://www.vals.ai/benchmarks/hlab-v1. Archived v1 protocol; GLM Legal and GLM 5.3 Flash task pass rates; accessed 25 October 2026.

[119] Ant Group. Ling-3.0-flash-Fin model card, September 2026. URL https://huggingface.co/inclusionAI/Ling-3.0-flash-Fin/blob/a5b4afb48f526175991724472c179e5a065604de/README.md. September 2026 release; evaluation chart and protocol in the official model card; reviewed 25 October 2026.

[120] Google. MedGemma 1.5 model card, January 2026. URL https://developers.google.com/health-ai-developer-foundations/medgemma/model-card. Official documentation of the 4B multimodal model, training inputs, and medical tasks; accessed 25 October 2026.

[121] Peter Brodeur, Jacob M. Koshy, Anil Palepu, et al. A prospective clinical feasibility study of a conversational diagnostic AI in an ambulatory primary care clinic, 2026. URL https://arxiv.org/abs/2603.08448v2. Cited version: v2, 15 March 2026.

[122] Niko Grupen, Julio Pereyra, Gabe Pereyra, Charlie O'Neill, Mudith Jayasekara, Aaron Ellis-Bloor, Jonathon Liu, and Matthew Blau. Post-training RLM agents for end-to-end M&A diligence. Harvey Research blog, September 2026. URL https://www.harvey.ai/blog/post-training-rlm-agents-for-m-and-a-diligence. Technical blog, September 8, 2026. Accessed October 19, 2026.

[123] Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to Route LLMs with Preference Data, 2024. URL https://arxiv.org/abs/2406.18665v1. Cited version: v1.

[124] Harvey. Security and data privacy, 2026. URL https://www.harvey.ai/security. Official hosting, regional processing, access, retention, and model-provider terms; accessed 25 October 2026.

[125] Google. MedGemma: Health AI developer foundations. Developer documentation, 2026. URL https://developers.google.com/health-ai-developer-foundations/medgemma. Updated January 13, 2026. Accessed October 23, 2026.

[126] Henry Mintzberg. Structure in 5's: A synthesis of the research on organization design. *Management Science*, 26(3):322–341, 1980. doi: 10.1287/mnsc.26.3.322.

[127] Mark Bovens. Analysing and assessing accountability: A conceptual framework. *European law journal*, 13(4):447–468, 2007. doi: 10.1111/j.1468-0386.2007.00378.x.

[128] Sumin Yu and Taesup Moon. Who will become the next senior? how generative AI erodes the development pathway in software engineering. arXiv preprint arXiv:2607.17067, 2026. URL https://arxiv.org/abs/2607.17067v3. Version 3, August 25, 2026. Accepted to AIES 2026.

[129] Fasheng Xu, Jing Hou, Wei Chen, and Karen Xie. Generative AI and organizational structure in the knowledge economy. arXiv preprint arXiv:2506.00532, 2025. URL https://arxiv.org/abs/2506.00532v2. Version 2, March 24, 2026; first submitted May 31, 2025.

[130] Seyed M. Hosseini and Guy Lichtinger. Generative AI as seniority-biased technological change: Evidence from U.S. résumé and job posting data. Working paper, SSRN 5425555, 2025. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5425555. Version October 6, 2025; first version August 31, 2025.

[131] Erik Brynjolfsson, Bharat Chandar, and Ruyu Chen. Canaries in the coal mine? six facts about the recent employment effects of artificial intelligence. Working paper, Stanford Digital Economy Lab, 2026. URL https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/. Revised August 12, 2026.

[132] Manuel Hoffmann, Sam Boysel, Frank Nagle, Sida Peng, and Kevin Xu. Generative AI and the nature of work. Harvard Business School Working Paper No. 25-021, 2025. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5007084. Version April 18, 2025.

[133] Tania Babina, Anastassia Fedyk, Alex X. He, and James Hodson. Firm investments in artificial intelligence technologies and changes in workforce composition. NBER Working Paper No. 31325, 2023. URL https://www.nber.org/papers/w31325. Issued June 2023.

[134] Stella Pachidi, Hans Berends, Samer Faraj, and Marleen Huysman. Make way for the algorithms: Symbolic actions and change in a regime of knowing. Organization Science, 32(1), 18–41, 2021. URL https://doi.org/10.1287/orsc.2020.1377.

[135] Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. The Quarterly Journal of Economics, 140(2), 889–942, 2025. URL https://doi.org/10.1093/qje/qjae044.

[136] Diana E. Forsythe. Engineering knowledge: The construction of knowledge in artificial intelligence. Social Studies of Science, 23(3), 445–477, 1993. URL https://doi.org/10.1177/030631279302300302.

[137] Sarah Lebovitz, Natalia Levina, and Hila Lifshitz-Assaf. Is AI ground truth really true? the dangers of training and evaluating AI tools based on experts' know-what. MIS Quarterly, 45(3), 1501–1525, 2021. URL https://doi.org/10.25300/MISQ/2021/16564.

[138] David Gray Widder and Dawn Nafus. Dislocated accountabilities in the "AI supply chain": Modularity and developers' notions of responsibility. Big Data & Society, 10(1), 2023. URL https://doi.org/10.1177/20539517231177620.

[139] Madeleine Clare Elish. Moral crumple zones: Cautionary tales in human-robot interaction. Engaging Science, Technology, and Society, 5, 40–60, 2019. URL https://doi.org/10.17351/ests2019.260.

[140] Inioluwa Deborah Raji, Andrew Smart, Rebecca N. White, Margaret Mitchell, Timnit Gebru, Ben Hutchinson, Jamila Smith-Loud, Daniel Theron, and Parker Barnes. Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency, 33–44, 2020. URL https://doi.org/10.1145/3351095.3372873.

[141] Mohammad Hasan Seifdar and Babak Amiri. Strategic adoption of generative AI in organizations: A game-theoretic and network-based approach. International Journal of Information Management, 84, 102932, 2025. URL https://doi.org/10.1016/j.ijinfomgt.2025.102932.

[142] Xiaoqing Wang, Wanle Zhong, Keman Huang, and Bin Liang. High interest but low adoption: Navigating organizations' journey towards generative artificial intelligence implementation. International Journal of Information Management, 87, 103009, 2026. URL https://doi.org/10.1016/j.ijinfomgt.2025.103009.

[143] Alex Singla, Alexander Sukharevsky, Bryce Hall, Lareina Yee, Michael Chui, and Tara Balakrishnan. The state of AI in 2025: Agents, innovation, and transformation. Survey report, McKinsey & Company, 2025. URL https://www.mckinsey.com/~/media/mckinsey/business%20functions/quantumblack/our%20insights/the%20state%20of%20ai/november%202025/the-state-of-ai-2025-agents-innovation_cmyk-v1.pdf. November 2025 edition.

[144] Sherwin Rosen. Learning and experience in the labor market. The Journal of Human Resources, 7(3), 326–342, 1972. URL https://doi.org/10.2307/145087.

[145] Hubert L. Dreyfus and Stuart E. Dreyfus. *Mind over Machine: The Power of Human Intuition and Expertise in the Era of the Computer*. Free Press, New York, 1986.

[146] Gary S. Becker. Human capital: A theoretical and empirical analysis, with special reference to education. Book, National Bureau of Economic Research and Columbia University Press,

New York, 1964. URL https://www.nber.org/books-and-chapters/human-capital-theoretical-and-empirical-analysis-special-reference-education-first-edition. First edition.

[147] Samuel Muehlemann and Stefan C. Wolter. Firm-sponsored training and poaching externalities in regional labor markets. Regional Science and Urban Economics, 41(6), 560–570, 2011. URL https://doi.org/10.1016/j.regsciurbeco.2011.04.003.

[148] Luis Garicano and Luis Rayo. Training in the age of AI: A theory of career viability. CEPR Discussion Paper No. 20634, 2025. URL https://cepr.org/publications/dp20634.

[149] Matthew Beane. Shadow learning: Building robotic surgical skill when approved means fail. Administrative Science Quarterly, 64(1), 87–123, 2019. URL https://doi.org/10.1177/0001839217751692.

[150] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. Generative AI without guardrails can harm learning: Evidence from high school mathematics. Proceedings of the National Academy of Sciences, 122(26), e2422633122, 2025. URL https://doi.org/10.1073/pnas.2422633122.

[151] Judy Hanwen Shen and Alex Tamkin. How AI impacts skill formation. arXiv preprint arXiv:2601.20245, 2026. URL https://arxiv.org/abs/2601.20245v2. Version 2, February 1, 2026.

[152] Daniel Kahneman and Gary Klein. Conditions for intuitive expertise: A failure to disagree. *American Psychologist*, 64(6):515–526, 2009. doi: 10.1037/a0016755.

[153] Enrique Ide. Automation, AI, and the intergenerational transmission of knowledge. CEPR Discussion Paper No. 20940, 2025. URL https://cepr.org/publications/dp20940.

[154] Zsófia L. Bárány and Miklós Koren. Broken ladders: AI, teamwork, and the dynamics of skill formation in the workplace. Working paper, 2026. URL https://koren.mk/publications/broken-ladders.

[155] Michael Link. When AI absorbs the apprenticeship: Rethinking staff development and expertise formation in survey research. Survey Practice, 20, 2026. URL https://doi.org/10.29115/SP-2026-0023.

[156] Margaret Stevens. Should firms be required to pay for vocational training? The Economic Journal, 111(473), 485–505, 2001. URL https://doi.org/10.1111/1468-0297.00647.

[157] OECD. SME technology adoption in the United Kingdom. OECD Studies on SMEs and Entrepreneurship, OECD Publishing, Paris, 2026. URL https://doi.org/10.1787/5f25ce2a-en. Published April 22, 2026.

[158] Claudia E. Haupt. Governing A.I.'s professional advice. McGill Law Journal, 64(4), 665–682, 2019. URL https://lawjournal.mcgill.ca/article/governing-a-i-s-professional-advice/.

[159] Claudia Haupt. Misplaced trust in artificial professional advice. Journal of Science & Technology Law, 32(1), 2026. URL https://scholarship.law.bu.edu/jostl/vol32/iss1/4/.

[160] Colleen V. Chien and Miriam Kim. Generative AI and legal aid: Results from a field study and 100 use cases to bridge the access to justice gap. Loyola of Los Angeles Law Review, 57(4), 2025. URL https://digitalcommons.lmu.edu/llr/vol57/iss4/2/. Starting page 903.

[161] Ira P. Robbins. Against an AI privilege. Harvard Journal of Law & Technology Digest, 2025. URL https://jolt.law.harvard.edu/digest/against-an-ai-privilege.

[162] Xiang Hui, Oren Reshef, and Luofeng Zhou. The short-term effects of generative artificial intelligence on employment: Evidence from an online labor market. Organization Science, 35(6), 1977–1989, 2024. URL https://doi.org/10.1287/orsc.2023.18441.

[163] Gueorgui Kambourov and Iourii Manovskii. Occupational specificity of human capital. International Economic Review, 50(1), 63–115, 2009. URL https://doi.org/10.1111/j.1468-2354.2008.00524.x.

[164] Xi Song, Jennie E. Brand, Sukie Xiuqi Yang, and Michael Lachanski. AI-accelerated occupational decline and the mobility trap. AEA Papers and Proceedings, 116, 246–250, 2026. URL https://doi.org/10.1257/pandp.20261004.

[165] Anton Korinek and Jai Vipra. Concentrating intelligence: Scaling and market structure in artificial intelligence. Economic Policy, 40(121), 225–256, 2025. URL https://doi.org/10.1093/epolic/eiae057.

[166] Jared Kirui. Financial services executive? Open AI wants you. News article, Finance Magnates, 2025. URL https://www.financemagnates.com/trending/financial-services-executive-open-ai-wants-you/. Reporting on OpenAI's Project Mercury, citing Bloomberg.

[167] Lin Kyi, Amruta Mahuli, M. Six Silberman, Reuben Binns, Jun Zhao, and Asia J. Biega. Governance of generative AI in creative work: Consent, credit, compensation, and beyond. Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems, 2025. URL https://doi.org/10.1145/3706598.3713799.

[168] Martin Senftleben. Generative AI and author remuneration. IIC – International Review of Intellectual Property and Competition Law, 54, 1535–1560, 2023. URL https://doi.org/10.1007/s40319-023-01399-4.

[169] Gordon Burtch, Dokyun Lee, and Zhichen Chen. The consequences of generative AI for online knowledge communities. Scientific Reports, 14, Article 10413, 2024. URL https://doi.org/10.1038/s41598-024-61221-0.

[170] R. Maria del Rio-Chanona, Nadzeya Laurentsyeva, and Johannes Wachs. Large language models reduce public knowledge sharing on online Q&A platforms. PNAS Nexus, 3(9), pgae400, 2024. URL https://doi.org/10.1093/pnasnexus/pgae400.

[171] Daron Acemoglu, Dingwen Kong, and Asuman Ozdaglar. AI, human cognition and knowledge collapse. NBER Working Paper No. 34910, 2026. URL https://www.nber.org/papers/w34910. Issued February 2026.

[172] Olle ten Cate. Entrustability of Professional Activities and Competency-Based Training. *Medical Education*, 39(12):1176–1177, 2005. doi: 10.1111/j.1365-2929.2005.02341.x.

[173] Fritz Machlup. *The Production and Distribution of Knowledge in the United States*. Princeton University Press, Princeton, NJ, 1962.

[174] Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. GPTs are GPTs: Labor Market Impact Potential of LLMs. *Science*, 384(6702):1306–1308, 2024. doi: 10.1126/science.adj0998.

[175] Ikujiro Nonaka. A dynamic theory of organizational knowledge creation. *Organization Science*, 5(1):14–37, 1994. doi: 10.1287/orsc.5.1.14.

[176] Andrew Abbott. *The System of Professions: An Essay on the Division of Expert Labor*. University of Chicago Press, Chicago, 1988.

[177] Herbert A. Simon. The structure of ill structured problems. *Artificial Intelligence*, 4(3–4): 181–201, 1973. doi: 10.1016/0004-3702(73)90011-8.

[178] Subbarao Kambhampati. Polanyi's Revenge and AI's New Romance with Tacit Knowledge. *Communications of the ACM*, 64(2):31–32, 2021. doi: 10.1145/3446369.

[179] Dhruv Saini. CAD-bench: Benchmarking Language Models on Functional CAD Generation. In *The Second Workshop on Agents in the Wild: Safety, Security, and Beyond (AIWILD), ICML 2026*, 2026. URL https://icml.cc/virtual/2026/67851.

[180] Afra Feyza Akyürek et al. PRBench: Large-Scale Expert Rubrics for Evaluating High-Stakes Professional Reasoning, 2025. URL https://arxiv.org/html/2511.11562v1.

[181] Qianyu Yang et al. $OneMillion-Bench: How Far are Language Agents from Human Experts?, 2026. URL https://arxiv.org/html/2603.07980v1.

[182] Burr Settles. Active learning literature survey. Technical Report 1648, University of Wisconsin–Madison, Department of Computer Sciences, 2009. URL https://minds.wisconsin.edu/handle/1793/60660.

[183] James A. Hanley and Abby Lippman-Hand. If nothing goes wrong, is everything all right? Interpreting zero numerators. *JAMA*, 249(13):1743–1745, 1983. doi: 10.1001/jama.1983.03330370053031.

[184] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback, 2022. URL https://arxiv.org/abs/2212.08073.

[185] Mint-Agent Team, Kun Wang, Gavin Zhang, Yaze Geng, Lei Tang, Yaoyang Yi, Zonghan Wu, Yifan Hu, Qingsong Wen, and Yilei Shao. Mint-Agent: Introducing Finance-Native Agentic Foundation Models, 2026. URL https://arxiv.org/abs/2608.16386v2. Cited version: arXiv:2608.16386v2, revised 21 August 2026.

[186] Raja Parasuraman and Dietrich H. Manzey. Complacency and bias in human use of automation: An attentional integration. *Human Factors*, 52(3):381–410, 2010. doi: 10.1177/0018720810376055.

[187] Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631 (8022):755–759, 2024. doi: 10.1038/s41586-024-07566-y.

[188] Vasudha Rengarajan, Karl de la Roche, Stephen Rice, Niko Grupen, Julio Pereyra, Gabe Pereyra, Nic Becker, Jacob Phillips, and Rhythm Garg. Training frontier Review Table models with Applied Compute. Harvey Research blog, August 2026. URL https://www.harvey.ai/blog/training-frontier-review-table-models-with-applied-compute. Technical blog, August 14, 2026. Accessed October 19, 2026.

[189] Ant Group. Ling-3.0-flash-Sante model card, October 2026. URL https://huggingface.co/inclusionAI/Ling-3.0-flash-Sante/blob/f74bd27bd26ffc71a50dd847805f8f7666ff0b43/README.md. October 2026 release; model card revision of 22 October 2026; evaluation table and local serving configuration.

[190] Supreme People's Court of the People's Republic of China. Interpretation on issues concerning the application of law in the trial of labour dispute cases (II). Judicial interpretation, Fa Shi [2025] No. 12, in Gazette of the Supreme People's Court, 2025, No. 10, 2025. URL http://gongbao.court.gov.cn/Details/bb72019c45453f84d920bd6375573e.html. Chinese text; full text of all 21 articles, including Articles 13, 14, and 21. Effective September 1, 2025. Accessed November 6, 2026.

[191] Supreme People's Court of the People's Republic of China. Interpretation on several issues concerning the application of law in the trial of labour dispute cases (IV). Historical judicial interpretation, Fa Shi [2013] No. 4, 2013. URL https://www.court.gov.cn/fabu/xiangqing/5064.html. Chinese text; Article 6. Repealed effective January 1, 2021 by Fa Shi [2020] No. 16, annex item 65. Accessed October 30, 2026.

[192] Standing Committee of the National People's Congress. Labour contract law of the people's republic of china. Full text published by the Beijing Municipal People's Government, 2012. URL https://www.beijing.gov.cn/zhengce/zhengcefagui/qtwj/201307/t20130701_780700.html. Chinese text, amended December 28, 2012; Articles 23–24. Accessed November 5, 2026.

[193] Supreme People's Court of the People's Republic of China. Decision repealing certain judicial interpretations and related normative documents. Fa Shi [2020] No. 16, 2020. URL https://www.court.gov.cn/fabu/xiangqing/282061.html. Chinese text; annex item 65, verified in the published image of the repeal list. Effective January 1, 2021. Accessed October 30, 2026.

[194] Supreme People's Court of the People's Republic of China. Interpretation on issues concerning the application of law in the trial of labour dispute cases (I). Judicial interpretation, Fa Shi [2020] No. 26, 2020. URL https://www.court.gov.cn/fabu/xiangqing/282121.html. Chinese text; Article 36. Effective January 1, 2021; read alongside the 2025 interpretation. Accessed October 30, 2026.

[195] Standing Committee of the Shenzhen Municipal People's Congress. Shenzhen special economic zone regulations on protection of enterprise technical secrets. Shenzhen Municipal People's Government Gazette, 2020, No. 34, 2019. URL https://www.sz.gov.cn/zfgb/2020/gb1164/content/post_8047987.html. Chinese text, amended October 31, 2019; Articles 2 and 24. Used as an out-of-territory candidate in the Suzhou example. Accessed November 5, 2026.

[196] Standing Committee of the Jiangsu Provincial People's Congress. Jiangsu labour contract regulations. Full text published by the Jiangsu Provincial Department of Justice, 2013. URL https://sft.jiangsu.gov.cn/art/2015/5/18/art_71926_8103093.html. Chinese text; Articles 2 and 28. Revised January 15, 2013; effective May 1, 2013. Accessed November 5, 2026.

A Author Contributions

Core Contributors. Yi Jing[1], Yudong Wang, Qibin Yang, Yixin Yang.

Contributors[2]. Jie Cao, Xinyi Dai, Ye He, Yiran Hu, Lei Li, Rang Li, Xinyu Li, Zhoufu Liu, Yuxuan Su, Zichao Tang, Wenqing Wang, Linghai Win, Miaoli Wu, Amy Xin, Weimin Xiong, Yuchen Yao, Xiao Zhang.

Advisors. Zhifang Sui[3], Juanzi Li, Sujian Li.

B Scope and Terminology

Boundaries of Knowledge Work. Our definition follows Drucker, who introduced knowledge work to set work whose principal resource is knowledge apart from manual labor [6, 18]; Machlup soon measured its share of the economy [173]. Knowledge work is a property of tasks more than of occupations, and exposure to AI is measured task by task [174]: an engineer reviewing a design and an analyst framing a valuation both do knowledge work, whether or not their occupation is licensed. The definition in Section 2 includes routine cognitive work in Autor's sense [19], such as bookkeeping and form processing, which explicit rules fully specify; it sits at the codified end as rule-based processing (Section 2.1). We focus on products carried in language, numbers, and digital artifacts, leaving aside expertise exercised through the body.

Kinds and Codification of Knowledge. Section 2.1 sorts three kinds of knowledge, factual, procedural, and experiential, by how far they have been or can be written down. The terms factual and procedural follow the knowledge dimension of the revised Bloom taxonomy [22], which echoes Ryle's distinction between knowing that and knowing how [21]. The three degrees of codification follow Cowan, David, and Foray's distinction between codified knowledge, knowledge that is unarticulated but articulable, and knowledge that cannot be articulated [23]. They also map onto Collins's kinds of tacit knowledge [24]: his *relational* knowledge, insofar as it could be written down, is our codifiable knowledge; his *somatic* knowledge, grounded in the body and appearing in knowledge work as perception, and his *collective* knowledge, which resides in a community, make up our tacit knowledge. Knowledge management more often speaks of explicit knowledge [175]; we use *codified* because what matters for AI is whether knowledge, and the success criteria it supports, have actually been written down.

Scientific Research. We treat scientific research as one domain of knowledge work, alongside the professions (Section 2). Choosing a question frames a task, designing a study and analyzing data execute it, and reviewing a manuscript judges it; our catalog includes such tasks, as when PaperBench asks for a published paper to be replicated [29]. What sets research apart is its product. A finding is a general claim rather than a result for a particular case, and whether it holds is settled by a research community over time. Peer review stands in for that verdict, as peer judgment stands in for late outcomes in the professions (Section 2.2), so the three stages and the role of tacit knowledge carry over; the concept of tacit knowledge was itself developed from scientific practice [16, 24]. We do not treat autonomous discovery systems separately. For them,

[1]Project lead.
[2]Contributors are listed alphabetically by last name.
[3]Corresponding author: szf@pku.edu.cn.

judging whether a claim is both new and true is the judgment stage in its hardest form, settled by a community over years, and how to evaluate it remains open.

Software as a Reference Point. Software engineering meets the definition: deciding what a system should do, choosing an architecture, and reviewing code all require framing and judgment. It is unusually closed, however. Its context lives in the repository and its outputs can be executed, so its execution stage is the most codified in knowledge work and can be checked against outcomes, which is why agents could be trained there first. We exclude it from the benchmark catalog and use it as a reference point (Section 3). Passing tests checks execution, but not whether the requirements were the right ones or whether a reviewer would accept the change.

Related Accounts of Expertise. The three stages are close to the classic account in the sociology of professions of professional work as diagnosis, inference, and treatment [176]. Simon observed that a problem is ill structured whenever a very large store of potentially relevant knowledge could bear on it [177]; LLMs bring such a store to every task and so meet this condition in extreme form, which is why framing is hard for them even when recall is easy. Dreyfus and Dreyfus describe how the novice who follows rules becomes the expert who sees what a situation calls for [145], the transition that peripheral work supports. Kambhampati uses Polanyi's revenge to describe the difficulty of guiding data-driven systems with explicit human knowledge [178]. Our second Polanyi paradox concerns the difficulty of articulating the success criteria needed to supervise these systems.

Use of Tacit Knowledge. Our use of the term is narrower than Polanyi's. For Polanyi, tacit knowing is the structure of all knowing, in which we attend from subsidiary clues to a focal whole, and all knowledge is either tacit or rooted in tacit knowledge [1]. We use the term, as studies of expertise and organizations do, for the part of expert experience that has not been, and cannot readily be, stated as general rules [23, 24]. Knowledge management has often treated such knowledge as convertible into explicit form; Tsoukas argues that it cannot be captured or converted, only displayed in skilled performance and refocused through dialogue about particular cases [25]. Our usage is compatible with this view: an expression of tacit knowledge records a judgment without the knowledge behind it, which is why we propose that agents learn from such expressions rather than wait for experts to write the knowledge down.

Knowledge in LLM Research. Section 2.1 contrasts the factual knowledge that LLM research has largely studied, facts a model stores in its parameters or retrieves on demand [8, 10], with the expert experience that tacit knowledge names. Two further points separate our usage. *Tacit* describes what experts can articulate, not what models encode; knowledge that a model holds but does not verbalize is a separate question. And the tacit knowledge that matters for knowledge work concerns success criteria, such as what matters in a case and what counts as good work, more than facts.

C Evaluation Coding Details

This appendix documents the coding behind Section 3.

Attributing the Five Decisions. The brief is coded as the AI actually receives it: the prompt, attached files, and standing instructions. Each of the five decisions (goal, materials, method,

Group	Benchmarks or subsets
Create (7 subsets)	Literary and advertising writing in WritingBench [41]; three design categories in RLI [42]; the arts category of JobBench [43]; the vague-prompt condition of UniPPTBench [44]
Heavy brief: method or conclusion fixed by the principal	ALE [48]; CAD-Bench [179]; SkillsBench [104]; JobBench [43]; investment-banking and consulting tracks of APEX-Agents [5]; most GDPval sectors [4]
Light brief: method and conclusion left to the AI	Analysis, drafting, review, and research tracks of Harvey LAB [27]; UniPPTBench [44]; PaperBench [29]; DSBench modeling [47]; OS-Marathon [46]; Tax Agent Bench [26]; HealthAgentBench [39]
Assist: professional question answering	PRBench [180]; $OneMillion-Bench [181]; Vals finance and legal suites [26]. Of the 50 *Assist* subsets, 36 are single-step tasks (L1).

Table 4. Benchmarks and subsets behind the groups discussed in Section 3.2.

structure, and substantive conclusion) is attributed to the principal, to professional convention, or to the AI. Choices that professional convention would dictate anyway are not counted as the principal's. Grading rubrics hidden from the model do not count as instructions, while rules supplied in attachments, knowledge bases, or operating policies count as a method set by the principal. *Collaborate* covers two situations: a brief that fixes decisions in advance, and a real person who intervenes during the work. Only Collaborative Gym [64] and CollabSkill [65] involve the second; simulated users and colleagues are not counted as humans in the loop.

Procedure. Subsets follow the categories each benchmark publishes. Some leaderboard pages list overlapping evaluation dimensions rather than a partition; we keep these as published. We read about 1,200 public task prompts across the catalog. When tasks within a subset fall at different levels (78 subsets), we assign the majority level and break ties by where the hard part of the task lies. Twenty-two subsets have no public tasks and are coded from papers or leaderboard descriptions; we flag them as low confidence. Prompts were read with LLM assistance, and every editorial change to a reading is recorded with its reason; we have not measured inter-coder agreement.

Weighting. Counts are reported per subset, which gives large suites more weight: ALE [48] contributes 14 subsets and GDPval [4] nine. With each benchmark weighted equally, the automation distribution becomes about 39% *Collaborate*, 36% *Assist*, and 24% *Delegate*; *Collaborate* remains the largest group.

Subsets Referenced in the Main Text. Table 4 lists the benchmarks and subsets behind the groups summarized in Section 3.2.

D Training Evidence and Validation Details

D.1 Repository-Based Task Synthesis

Coding research provides concrete construction mechanisms for Section 4.1. CodeMidas selects functionality from existing code, removes key implementation components, and builds a task description and hidden tests around reference behavior. Cleanup, execution checks, and rollout-based review help establish that the task and verifier agree [70]. SWE-smith instead prepares a reusable repository environment and generates many repair tasks within it, amortizing setup

Table 5. Training in professional workflow environments. Results use different models, task splits, and success criteria and are not directly comparable.

Study	Experience source	Training evidence and scope
EnterpriseLab [74]	Enterprise applications and executable tool-dependency paths.	SFT, DPO, and GRPO with data constructed separately for each benchmark.
Corecraft [75]	A coherent enterprise state and expert-authored tasks.	GLM-4.6 with GRPO: held-out full-task pass rate rises from 25.37% to 36.76%; external transfer is also reported.
HealthAdminBench [76]	EHR, payer portals, and fax workflows.	Qwen-3.5-27B with LoRA SFT: 100 training tasks and 35 held-out tasks; task success rises from 17.1% to 40.0%.
Spreadsheet-RL [77]	Forum problems converted into initial and target workbooks.	Qwen3-4B-Thinking with RL: SpreadsheetBench Pass@1 rises from 19.3% to 23.4% with the full tool interface held fixed.
BankerToolBench [78]	Financial models, presentations, and memos.	Dr. GRPO and DPO on frozen rollout pools with an 80/20 task split; an offline post-training experiment.

across instances [14]. These methods illustrate how executable systems can supply both problems and evidence about their solutions. For knowledge work, the transferable design question is which business functions and constraints admit similarly repeatable, checkable interactions.

D.2 Validation of Generated Environments and Simulators

ScaleEnv and AutoForge construct tools, databases, and tasks from domain descriptions or tool documentation, using execution to check reference solutions [71, 72]. Reference-solution success establishes feasibility but does not establish correctness across the action space. An environment that accepts only the reference path can penalize valid alternatives, so validation should test alternative solutions as well as implementation semantics. DreamGym's experience model predicts subsequent states and rewards from real offline trajectories [73]; learned simulators additionally require validation of prediction error and transfer to real execution.

D.3 Training Evidence in Professional Workflows

Table 5 records representative training experiments. The studies use different models, task splits, and success criteria, so their gains are not directly comparable. Low-rank adaptation (LoRA) denotes parameter-efficient fine-tuning.

Enterprise experiments. EnterpriseLab builds dependency graphs over enterprise tools, samples executable sequences, and generates user intents for SFT, direct preference optimization (DPO), and online group relative policy optimization (GRPO). It constructs training data separately for each benchmark, so its results across environments demonstrate platform adaptability without establishing zero-shot transfer from a single training environment [74]. Corecraft combines a coherent customer-support organization with expert-authored tasks and rubrics, and reports both held-out full-task success and external transfer [75].

Stage	Mechanism	Representative evidence	Status
Seed	Professional questions, procedures, knowledge relations, or observed practice constrain generation.	Fin-o1 [79]; LegalOne [80]; MedReason [81]; BRIDGE [82].	Reported
Generate and filter	Models generate reasoning, check outputs, and repair failures.	STaR [83]; Agentar [84]; MedReason [81].	Reported
Review and feed back	Expert judgments change subsequent generation or training.	MedReason: review only [81]. PRM800K and ILF: feedback outside the professional settings reviewed [85, 86].	Partial
Expand and audit	Coverage grows, targeted review finds failures, and random audits bound rejection.	Active learning and sampling bounds inform the proposed protocol [182, 183].	Proposal

Table 6. An organizing framework for expert-seeded synthesis. "Reported" indicates professional-domain instances of the stage; "Partial" separates expert evaluation from feedback used in training; "Proposal" identifies the auditing cycle developed here. The table does not imply that any single study implements the complete cycle.

Administrative and artifact-based workflows. HealthAdminBench covers prior authorization, denial appeals, and equipment orders, which require transferring information and maintaining progress across systems. Its small SFT experiment supports workflow adaptation under the tested observation and prompting configuration, not a general scaling law [76]. Spreadsheet-RL supplies execution feedback by recalculating edited workbooks [77]. BankerToolBench's post-training experiment uses frozen rollout pools, showing that its grading signal can guide optimization; the released benchmark remains an evaluation resource rather than evidence of online environment scaling [78].

D.4 Evidence and Reporting for Effective Scale

In EnterpriseLab's fixed setting, increasing the training set from 500 to 1,000 and then 1,500 instances does not yield monotonic gains [74]. This motivates checking coverage and redundancy; it does not identify their separate causal effects. Adding instructions within a narrow workflow and adding a new policy constraint or system dependency are different forms of expansion. Useful comparisons should state which task and state dimensions change and test whether feedback distinguishes the behaviors produced by the current policy.

Cost includes environment construction, resets, retries, grading, and expert review. A cheap simulator can reduce collection expense, but its business rules and task distribution still need calibration against intended work. Reporting validated interactions, coverage, cost, and independent transfer alongside task counts distinguishes operational scale from evidence of improved performance.

D.5 Evidence for Expert-Seeded Synthesis

Table 6 organizes the evidence reviewed in Section 4.2. It distinguishes mechanisms documented in the cited studies from the feedback and auditing protocol proposed here; its status labels do not assert the absence of other professional-domain research.

Generation and filtering experiments. Fin-o1 uses expert-derived subquestions and program steps; GPT-4o generates reasoning and another GPT-4o instance verifies and revises it, producing 9,186 CoT-annotated pairs [79]. LegalOne's LEAD pipeline uses expert protocols and knowledge

bases, then rewrites and merges legal trajectories [80]. MedReason constrains generation with medical knowledge-graph paths and retains 32,682 of 45,725 generated chains after checking whether a model recovers the reference answer. Its filtering ablation changes the average score from 0.539 to 0.550 on one benchmark suite [81]. This isolates a contribution under that configuration; it does not show that filtering validates professional reasoning or establish why the gain is small.

Expert review and articulation. MedReason reports physician comparisons with HuatuoGPT-o1 chains in seven specialties, using more than 25 randomly sampled chains per specialty. Physicians prefer MedReason in every specialty, unanimously in gastroenterology, but the reported preferences serve as evaluation rather than iterative generator feedback [81]. Cho et al. use 320 tutoring conversations containing novice and expert responses to have GPT-4o articulate decision steps. Generated tutoring outperforms that based on the three-step structure elicited through BRIDGE, and novices trained with the model's articulation come closest to expert performance in the reported comparison [31, 82]. The ratings come from an LLM judge, so this supports extracting a useful articulation without independently establishing expert endorsement.

Feedback precedents and adjacent mechanisms. PRM800K supplies 800,000 step-level human labels on competition-mathematics solutions, uses active learning to select solutions, and reports an advantage for process over outcome supervision [85]. Its labeling setting differs from professional review. Imitation learning from language feedback (ILF) turns human critiques into model-written refinements used as training targets [86]. Constitutional AI instead scales human-authored principles through model-generated critiques and revisions [184]. These are precedents for transmitting feedback, not demonstrations of a professional expert loop. Harvey diligence filters successful agent runs for self-distillation, while Mint-Agent supplies teacher guidance at critical states reached by the student [122, 185]. Their improvements likewise do not establish expert validation of synthesized reasoning.

D.6 Auditing and Scaling Synthetic Reasoning

The following protocol is a proposal for testing and extending the synthesis loop. It separates finding useful corrections from estimating how often a specified review protocol would reject generated material.

Targeted review and random audits. Active learning can prioritize uncertain or unfamiliar cases, including convincing but incorrect solutions, for informative feedback [85, 182]. Such selection does not directly estimate population rejection rates. A separate audit can use independent, identically distributed draws from a fixed generation distribution under a fixed review protocol. If none of n draws is rejected, the one-sided 95% binomial upper bound on rejection probability is $1 - 0.05^{1/n}$, approximately $3/n$ [183]. Thus, zero rejections among 100 or 1,000 draws gives approximate bounds of 3% or 0.3%. These bound review rejection, not every factual error: reviewers may miss defects. A revised generator changes the distribution, so a previous bound does not automatically validate the new round. Reports should specify sampling, exclusions, reviewer agreement, rejection counts, and review cost.

Review bias and distributional coverage. Human use of automated decision support is susceptible to complacency and automation bias [186]. Comparing alternatives and asking reviewers to identify missing evidence can make review more informative than approval alone. Recursive

training on generated data can also lose distributional tails [187]. External expert feedback is a potential corrective signal, not a guarantee against this failure. Reviewing batches as well as individual chains can reveal homogeneity, while tests on changed conditions and uncommon cases assess whether expansion preserves coverage.

A proposed comparison of supervision stages. On a common task set, an informative study would hold the training recipe and budget fixed while comparing answer-only supervision, synthesized chains, automatically filtered chains, expert-reviewed chains, and chains repaired after expert rejection. It should assess unseen work independently, including missing evidence and changed conditions, and report expert effort. LegalOne, Fin-o1, and MedReason report gains from combined recipes that do not isolate the expert-feedback stage [79–81]. This comparison therefore specifies evidence still needed for the proposed loop rather than an established ranking. Fluent teaching rationales need not faithfully expose a model's internal computation [87]; their training value must be assessed through resulting work.

D.7 Reward Construction and Expert Calibration

The following cases complement AI rubric synthesis in Section 4.3. They separate applying predefined professional criteria at scale from generating new criteria, and distinguish policy-training results from studies of evaluator alignment.

Applying professional criteria during training. Harvey Review Table takes the minimum of schema adherence and answer correctness as its base reward, then adjusts it using citation precision times recall. A wrong answer therefore cannot earn credit through citations alone [188]. This combines programmatic checks and model judgments around specified requirements; it does not establish automatic synthesis of task-specific rubrics. Harvey's diligence experiment instead uses an expert rubric with hundreds of criteria. With sub-agent models fixed, GRPO trains the root agent on judged criterion pass rate, raising the held-out rate from 29.9% to 63.0% across 50 data rooms [122]. The evidence concerns root-policy improvement, not joint optimization of every agent or the probability of an entirely acceptable memorandum. Corecraft likewise uses fractional criterion satisfaction during training while reporting a stricter all-criteria pass rate [75]. Dense rewards can guide learning when complete success is rare; final acceptance should still be measured separately. These examples show how expert-authored or explicitly designed standards become reusable rewards, rather than how AI generates the standards themselves.

Calibrating generated rubrics. Harvey's professional-preference study uses generative reward models that construct rubrics, compare outputs, and verify claims with tools. Legal researchers guide the rubric axes, and lawyers assess the resulting criteria and reasoning. Their disagreements expose inappropriate trade-offs, including compensating for serious hallucinations with stylistic advantages [90]. This supports expert calibration of generated judgments, but the reported study evaluates a judging protocol; it does not demonstrate policy RL driven by these generated rubrics. The same report states that Tenet's training used pre-authored expert rubrics. Transferring such calibration into a repeated synthesis-and-training cycle therefore remains a separate empirical step.

E Deployment Evidence and Procedural Examples

This appendix records the interventions, evaluation settings, and procedural details behind Section 5. The comparisons concern different units: adding a resource, revising a procedure, chang-

ing the runtime, selecting a model, or altering delegation. Keeping those units explicit helps connect each result to a deployment decision.

E.1 Procedural Support and Runtime Comparisons

SkillsBench v4: adding curated packages. The benchmark compares complete instruction-and-resource packages with no-Skill configurations on 87 terminal-workflow tasks across 18 model–harness configurations. It reports task-macro pass rates; the finance/economics group improves from 19.1% to 33.3%, and office work from 40.5% to 53.0%. Task selection includes measurable separation between conditions. Its length analysis compares groups of resources, and its negative-trajectory analysis identifies heavy pipelines, displacement of effective default approaches, and brittle dependencies [104].

BRIDGE: comparing tutoring replies with expert-written feedback. The task is to respond to a student's mathematical mistake. The procedure identifies the error, chooses a strategy, and specifies the intended teaching effect. For each response source, 40 randomly sampled reply pairs are rated by three teachers per pair, blind to the source. Scores range from -2 to 2 relative to the original tutor reply. GPT-4's preference means are 0.54 for direct generation and 1.02 when it selects its own decisions; expert-written replies score 1.26. A separate condition supplies expert decisions for the current case and scores 0.95. The paper reports significant preference and usefulness gains for GPT-4's self-selected procedure over direct generation [82].

AMIE (Video): physician comparison on standardized cases. The August 2026 study evaluates an integrated Talker–Planner–Perception system built with Gemini 3 Flash and Gemini 3.1 Pro. Fifteen patient actors enact 100 clinical scenarios across study arms, including AMIE video and 10 US board-certified primary-care physicians consulting by video. Twenty other physicians evaluate recordings, transcripts, and responses using general and case-specific rubrics. The aggregate case-specific score is 83% for AMIE video and 68% for physicians. Comparisons use scenario-level pairing with false-discovery-rate adjustment [105].

SkillRefine: revising the resource with tools held fixed. The library is constructed using documentation, expert edit records, and execution results from 40 construction tasks, then evaluated on 100 held-out tasks from two PIMS demonstration models. With DeepSeek-V4-Pro and the same tools, component-match F1 rises from 0.694 for the unrefined library to 0.770 after refinement. The metric measures agreement with reference edits to planning-model components [102].

Runtime effects and transfer. MedicalHarness evaluates five open-weight models under five harnesses on 107 executable tasks, keeping serving conditions fixed and repeating runs across seeds. Its roughly one-quarter variance share combines the harness main effect and model–harness interaction. FinanceHarness's fixed Qwen3.6-27B ablation reports normalized rubric means of 25.3% with search alone, 29.6% with a naive harness, and 32.4% with the full harness before additional policy training [112, 113]. Harness Scaling instead trains across 18 configurations and tests unseen combinations and external harnesses on workspace benchmarks; prompts, interfaces, observations, context management, and workflow are the varied dimensions [116].

E.2 Model Selection and Division of Labor

Recent domain-model comparisons. In the comparison of Section 5.4, each task in Vals' archived Harvey LAB v1 must satisfy all its criteria. The October 2026 release also reports

LegalBench scores of 86.2 for GLM-Legal, 83.9 for its GLM-5.3-Flash base, and 88.6 for Claude Fable 5 [117, 118]. These comparisons cover both complete legal tasks and legal reasoning under their respective protocols.

Ling-3.0-flash-Fin's September 2026 model card reports Finance Agent v2 scores of 59.81 for Ling-fin, 56.48 for GPT-5.6-Sol, and 64.20 for Gemini-3.7-Flash, using the official harness and dataset with strict-pass scores averaged over 20 runs. SpreadsheetBench v2 uses Claude Code and reports 21.81 for Ling-fin against 28.35 for Claude-Opus-5 [119]. The October 2026 Ling-3.0-flash-Sante card reports HealthBench Professional scores of 54.3 for Sante, 49.1 for GLM-5.3-Flash, and 60.8 for GPT-5.6-Sol, using the highest reasoning-effort tier for compared models. Its MedQA-USMLE scores are 94.2 and 95.2 for Sante and GLM-5.3-Flash, respectively [189].

Harvey diligence: changing model allocation and depth. In a comparison of four root models and three worker models on 30 synthetic held-out data rooms, the spread between the highest- and lowest-scoring roots averages about 38 percentage points with workers held fixed; the corresponding spread across workers averages about 8 points. A separate depth comparison uses Qwen3.5-122B-A10B for the root and all workers on 14 rooms. Allowing workers to use a REPL and delegate further lowers mean criteria pass rate by 19 percentage points; four regression cases read documents but produce no report [122]. This comparison changes worker tools as well as delegation depth.

E.3 Case Study: Designing a Skill for Financial Report Generation

Designing a Skill for financial report generation involves specifying how to acquire information, perform analysis, and decide whether a report is ready for use. We contrast a general workflow designed by technical staff without a finance background with one that incorporates financial professionals' methods and acceptance criteria. We first describe each reusable workflow, then apply both to an annual earnings commentary for a listed manufacturer. The company data and commentary excerpts are constructed examples. The comparison follows how the Skill's instructions shape the evidence collected, the reasoning performed, and the conclusions delivered.

E.3.1 A General Technical Workflow

The first design organizes report generation around information processing. It specifies a sequence of tool operations and document-production steps that can be reused across report topics.

1. **Understand the request.** Identify the report's topic, the information to collect, and the requested format. Translate the request into an outline and a list of retrieval and calculation tasks, such as finding the relevant documents and summarizing changes in key indicators.
2. **Search for material.** Use web search, page fetching, and document download tools to obtain relevant sources. Pass the collected text and files to the extraction stage once material matching the topic has been found.
3. **Extract information.** Parse the documents for numbers, facts, and stated opinions. Organize this material into fields or notes that can be supplied to calculation tools and the report-writing model.
4. **Calculate and summarize.** Use Python or Excel to calculate indicators, aggregate observations, and construct tables. These operations turn extracted values into concise numerical summaries for the report.

5. **Write the report.** Ask the model to produce analysis, conclusions, and citations from the collected material and computed indicators. Arrange the output under the requested headings and export the document.
6. **Run technical checks.** Confirm that retrieval and parsing completed, calculations executed correctly, references open, and the file contains the required sections. Check that the document's formatting is complete and readable before returning it.

This design makes the retrieval–calculation–writing sequence executable. Its instructions leave several analytical decisions to the model, including which financial measures are necessary, how to explain their changes, and what evidence supports an assessment of business performance. Those decisions become explicit in the second design.

E.3.2 A Workflow Incorporating Financial Expertise

Financial professionals retain the general workflow's tool operations and add rules for selecting data, interpreting results, and accepting the analysis. The Skill records these rules as required inputs, calculation conventions, questions to resolve, and checks that determine whether to revise or deliver a report.

1. **Understand the request and its audience.** Establish the report's topic, intended readers, information needs, and format. A reader assessing operating performance needs an account of the drivers and quality of earnings, which determines what the analysis must cover.
2. **Define the analytical scope.** Fix the company or other entity, reporting period, information cutoff, and questions to answer. Use these choices to filter sources and comparisons so that the report consistently addresses the same entity and period with information available at the specified time.
3. **Set source priorities.** Specify which sources serve each information need. Where access is authorized, services such as Wind or iFinD can supply standardized financial indicators. Company and exchange disclosures provide the original statements and explanations for verification. Resolve conflicting values by checking definitions, versions, and update times.
4. **Retrieve the required documents.** Collect reports, announcements, and relevant industry material through the available interfaces and search tools. Define the statements and notes that must be read for the task, check their availability, and record document versions before proceeding with analysis.
5. **Extract facts with their context.** Retain each value's source, table or page location, report date, and explanatory notes. Record management's explanations alongside the relevant figures. This preserves the connection between a number and the disclosure needed to interpret it.
6. **Align data definitions.** Reconcile units, consolidation scope, fiscal periods, and indicator definitions before comparison. Distinguish annual from quarterly values, resolve revised versions, and preserve the difference between a disclosed zero and a missing observation. Record any adjustments used to make values comparable.
7. **Calculate comparable indicators.** Use Python or Excel for growth rates, historical or peer comparisons, and other required measures. Retain input values, formulas, and the basis of comparison so that a reader can reproduce a result and trace it to the underlying data.
8. **Investigate the drivers of change.** Examine whether movements reflect sales volume, prices, costs, non-recurring gains, or other factors. Connect each proposed explanation to a disclo-

Metric	2024	2025
Revenue	10.0	11.2
Attributable net profit	1.0	1.5
Non-recurring profit/loss	0.0	0.6
Net profit excluding non-recurring items	1.0	0.9
Gross margin	20%	18%

Table 7. Illustrative annual-report data. Amounts are in RMB 100 million except gross margin. Profit items are after tax and attributable to parent shareholders.

sure or a supported calculation. Where a driver cannot be resolved, specify the additional information needed to assess it.

9. **Write the analysis from evidence.** Distinguish reported facts, management's explanations, and the analyst's inferences in the wording and citations. Match the strength of a conclusion to the supporting evidence, and connect unresolved questions to the particular information that remains missing.
10. **Run technical checks.** Verify tool execution, calculations, references, file integrity, and formatting. Ensure that tables and charts display correctly and that the delivered document and any accompanying workbook can be opened and used.
11. **Conduct professional verification.** Check that numbers are traceable, comparisons are meaningful, the reasoning supports the conclusions, and the prose agrees with the tables. Failed checks lead to additional retrieval, recalculation, or revision. Material judgments and unexplained anomalies are referred for human review within the current assignment.
12. **Deliver a standardized report.** Present the core conclusions, key indicators, explanations of changes, risks, and open questions in a consistent structure. Include sources and necessary calculation notes so that the recipient can inspect the reasoning and continue the analysis from the delivered work.

These instructions connect stages of the workflow. The scope determines which documents qualify, the required reading determines which facts are extracted, and the data definitions determine which calculations are meaningful. Professional verification can send a draft back to an earlier step when a conclusion requires evidence that has not yet been collected.

E.3.3 Worked Example: Annual Earnings Commentary

The task is to prepare a short annual earnings commentary for Company A, an illustrative listed manufacturing company. The request is:

Task Request

As of 30 April 2026, prepare an earnings commentary based on Company A's 2025 annual report.

Both workflows have access to the same annual report, published before the cutoff. Its disclosures include the figures in Table 7 and a note explaining that the non-recurring gain comes from selling an idle factory. The report provides insufficient volume, selling-price, and unit-cost detail for a complete decomposition of changes in revenue and gross margin.

Applying the general technical workflow. The six steps below follow the first Skill design. They show how a report can contain correct headline figures while its interpretation goes beyond

the analysis performed.

1. **Understand the request.** Classify the assignment as a commentary on Company A's 2025 results. Plan sections on revenue, profit, operating performance, and risks. The task becomes one of collecting the annual results and summarizing their changes.
2. **Search for material.** Search for Company A's 2025 annual report and retrieve the document. Once the annual report is available and readable, proceed to extract the figures used in the planned commentary.
3. **Extract information.** Read the headline financial indicators or report summary. Extract revenue of 11.2 and attributable net profit of 1.5 for 2025, together with the corresponding 2024 values of 10.0 and 1.0, all in RMB 100 million.
4. **Calculate and summarize.** Compute year-on-year changes as the current value divided by the prior-year value, minus one. Python or Excel returns revenue growth of 12% and attributable net-profit growth of 50%. Place these correct results in an indicator table.
5. **Write the report.** Use the two growth rates to draft a performance assessment. The Skill has not required a decomposition of profit or evidence for an inference about core operations. An illustrative draft reads:

   Commentary: General Workflow

   Company A's revenue rose 12% and attributable net profit rose 50% in 2025. These increases demonstrate **improved core operations and stronger profitability**.

6. **Run technical checks.** Confirm that the retrieval and parsing tools succeeded, the growth rates match the input values, the citation opens, and the required sections are present. The report passes these checks. The inference about core operations has not been tested against the composition of earnings.

The analytical gap lies between the calculated growth rates and the claim about business performance. The available annual report contains information needed to examine that claim, but the first design does not make reading or reconciling that information a condition of completion.

Applying the workflow with financial expertise. The twelve steps below apply the second design to the same request. Financial rules change both what the workflow collects and how it uses the collected information.

1. **Understand the request and its audience.** Treat the commentary as an explanation for a reader assessing annual performance. Ask where growth came from, how reported earnings relate to day-to-day operations, and which drivers require further investigation. These questions establish the purpose of the numerical analysis.
2. **Define the analytical scope.** Resolve the company identity and security identifier when locating disclosures. Set the comparison to full-year 2025 against full-year 2024 on a comparable consolidated basis. Apply the 30 April 2026 cutoff to document selection and use the versions disclosed by that date.
3. **Set source priorities.** Treat the original annual report as the source for the figures and their explanations. If an authorized Wind or iFinD connection is available, use standardized fields to facilitate retrieval and reconcile them with the report. Investigate differences in period, definition, or version before choosing a value.

4. **Retrieve the required documents.** Follow a required-reading list covering the income statement, non-recurring-items schedule, relevant notes, and operating discussion. Continue past the headline results to the note on the idle-factory disposal. This step makes the source of the non-recurring gain available to the analysis.

5. **Extract facts with their context.** Collect all five indicators in Table 7, the disposal explanation, and the associated source locations and notes. Record that sufficient volume, price, and unit-cost information is unavailable. The evidence record now includes both headline growth and the components needed to interpret it.

6. **Align data definitions.** Verify the full-year periods and comparable consolidation scope, and express amounts in RMB 100 million. Confirm that the 0.6 disposal gain is after tax and attributable to parent shareholders. Sale proceeds or a pre-tax disposal gain would require reconciliation before entering the same profit bridge.

7. **Calculate comparable indicators.** Obtain revenue growth of 12%, attributable net-profit growth of 50%, a 10% decline in net profit excluding non-recurring items, and a gross-margin decline of 2 percentage points. Decompose the net-profit increase into the change excluding non-recurring items and the change in non-recurring profit. In the table's units, the bridge is:

   $$0.5 = -0.1 + 0.6$$

   Retain the two years' inputs and formulas. The 0.6 increase in non-recurring profit more than accounts for the 0.5 increase in total attributable net profit, offsetting the 0.1 decline in the component excluding non-recurring items.

8. **Investigate the drivers of change.** Link the 0.6 increase to the disposal note. The factory sale explains the rise in net profit. For revenue and gross margin, request the volume, selling-price, and cost information needed to distinguish possible drivers. Lower adjusted profit and margin also prompt investigation of the operating picture, including product mix and expenses where relevant evidence is available.

9. **Write the analysis from evidence.** Report the revenue increase and attribute the net-profit increase to the factory-disposal gain. Present the decline in net profit excluding non-recurring items and in gross margin alongside those increases. Keep possible volume, price, and cost explanations as questions for investigation until the corresponding evidence is available.

10. **Run technical checks.** Recalculate the rates and the profit bridge, open the report and any workbook, verify references, and inspect the table. Check that percentages and percentage-point changes are labeled correctly and that the figures remain consistent after export.

11. **Conduct professional verification.** Test any claim of improved core operations against the disposal note, adjusted profit, and gross margin. If the draft still asserts broadly stronger profitability, require an explanation consistent with those measures. Revise unsupported language and identify any additional evidence or human review needed to resolve a material question before delivery.

12. **Deliver a standardized report.** Return a performance overview, indicator table, profit bridge, follow-up questions, and source and calculation notes. The commentary can read:

Commentary: Financial Expertise

Company A's revenue grew 12% and attributable net profit grew 50% in 2025. The profit increase was driven by an after-tax, attributable gain of RMB 60 million from selling an idle factory. Net profit excluding non-recurring items fell 10%, and gross margin declined by 2 percentage points to 18%. Further analysis of revenue and margin changes requires sales-volume, selling-price, and unit-cost information.

Where the Skill designs differ. The differences begin before calculation. The financial workflow turns the non-recurring-items schedule and disposal note into required reading, so the extracted data cover the composition of earnings. Its definition checks establish that the disposal gain and net profit can be reconciled on an after-tax, attributable basis. Its analytical rules then convert two growth rates into an explanation of why profit increased, while identifying the separate information needed to explain revenue and margin changes.

The designs also use different criteria for completion. The general workflow checks that the report was produced correctly from its selected inputs. The financial workflow additionally checks that those inputs and the analysis support the report's claims. In this example, financial expertise is expressed through required evidence, comparable definitions, a profit-decomposition method, and rules for revising unsupported conclusions.

E.4 Case Study: Designing a Skill for Statutory Retrieval

A Skill for statutory retrieval must connect a person's problem to the provisions that govern it. We compare a general retrieval workflow designed by technical staff without legal training with a workflow incorporating legal expertise. We first describe the two reusable workflows, then apply each step to a constructed employment scenario in Suzhou, China, using illustrative report excerpts and verified public legal sources, including judicial interpretations of the Supreme People's Court (SPC). The comparison follows how task understanding, workflow design, and error handling shape the report. Together, issue formulation, retrieval and application, and professional review connect the example to the three stages in Section 2.2.

E.4.1 A General Technical Workflow

The first design organizes legal research around retrieving relevant passages. It treats the request as a set of topics and uses query expansion, keyword and semantic search, reranking, and source verification. Its main checks concern coverage, relevance, and citation accuracy.

1. **Parse the request.** Extract the parties, place, dates, and topics, and identify the expected output as a list of legal provisions with explanations.
2. **Form queries.** Generate queries for each topic and expand their wording with related expressions. These queries increase the chance of finding provisions expressed differently from the user's question.
3. **Retrieve materials.** Search legal databases and government or court websites, retaining the provision number, document title, and source link for each result.
4. **Rank and organize.** Remove duplicate passages, rerank candidates by relevance, and record publication dates. Check that the original text can be obtained and that the quotation matches it.
5. **Repair retrieval problems.** Broaden searches for topics with few results, retry failed downloads or parsing, and replace irrelevant passages or incomplete citations.

6. **Produce the report.** Summarize the provisions under the requested topics. Verify that each topic is covered, citations resolve, and the report is complete and readable.

This design makes legal materials retrievable and the report traceable to its sources. It leaves the model to determine which conditions govern a particular question and how the supplied facts satisfy them. The second design makes those decisions explicit in the procedure.

E.4.2 A Workflow Incorporating Legal Expertise

The legally informed design organizes the same search capabilities around questions with conditions and dependencies. It first identifies the legal relationship and the issue to be decided. A person's job title, a signed agreement, and a relevant provision each supply part of the analysis. Their significance depends on the conditions of the rule and the facts of the assignment. Public statutory-retrieval procedures likewise connect legal relationships, source validity, application to facts, and renewed retrieval [103].

1. **Identify the legal relationship and requested determination.** Establish whether the question concerns employment, a corporate office, or another relationship. Separate restrictions during employment from those after departure, then identify the questions the report must answer.
2. **Build a fact record.** Record the parties' roles, relevant locations and dates, the agreement, the protected information actually accessed, and the requested relief or advice. Attach the available evidence to each fact and identify missing facts that could change the analysis.
3. **Decompose the issues.** Turn broad topics into questions about eligibility, scope, and consequences. Identify the conditions and exceptions for each question, including whether compensation has been agreed and which local rules govern the relationship.
4. **Retrieve by issue and source type.** Obtain the statutory framework, relevant judicial interpretations, and applicable local provisions. Search for exceptions and cross-referenced rules alongside affirmative provisions. Keep published cases identifiable as materials explaining application.
5. **Verify versions and authority.** Record the issuing body, document number, territorial scope, publication and effective dates, and any amendment or repeal. Follow an old provision to its current counterpart and retain the basis for the version decision.
6. **Check applicability to each issue.** Match a candidate's temporal, territorial, personal, and substantive conditions to the fact record. Mark it as usable for the issue, inapplicable to that issue, or pending a factual check, and record the reason.
7. **Investigate facts required by the rule.** When a provision introduces a decisive distinction, request the corresponding evidence. For a restriction tied to technical secrets, investigate the employee's access and the relationship between those secrets and the proposed business restriction. Revisit the analysis when the facts become clearer.
8. **Resolve apparent conflicts.** Determine whether competing passages govern the same facts and issue before applying rules about authority or precedence. Distinguish an agreed payment from a default for missing terms, and a local requirement from a provision for another territory.
9. **Deliver and review the reasoning.** Report the applicable provisions, their connection to the facts, the supported conclusions, and the questions still requiring investigation. Preserve important exclusion decisions so that review can check why a retrieved provision did or did not enter the answer.

These instructions make legal distinctions control the workflow. A missing fact holds up the judgment that depends on it while source verification can continue. A new provision may reveal a question the initial brief omitted, sending the system back to the fact record and then to a more precise search.

E.4.3 Worked Example: A Post-Employment Non-Compete Clause

Both workflows address the same request and supplied facts. The request is:

Task Request

As of 30 November 2025, identify the legal provisions governing employee Li's post-employment non-compete clause with a technology company in Suzhou. Explain who may be bound, how the restriction's scope should be examined, and how compensation should be determined. State the applicable conditions and any facts still needed.

The company is registered and operates in Suzhou, Jiangsu Province, and Li performs the employment contract there. Li is an algorithm testing engineer who has accessed unpublished compression parameters and deployment designs for a particular image-algorithm product. The example treats these materials as protected technical secrets, with access records confirming Li's exposure. The parties sign the clause on 15 September 2025, and Li leaves on 31 October 2025. The clause prohibits joining any business engaged in artificial intelligence anywhere in China for 12 months after departure.

The parties expressly agree on monthly compensation of 30% of Li's average monthly wage during the preceding 12 months. That verified wage is RMB 12,000, making the agreed payment RMB 3,600. The prospective employer's products, their competitive relationship with the protected product, and the justification for the clause's broad business and geographical scope remain to be established. Both the agreement and departure occur after the SPC's 2025 interpretation takes effect on 1 September 2025 [190].

Applying the general technical workflow. The six steps below follow the first design. They show how a report can quote genuine provisions while leaving their applicability to the case unchecked.

1. **Parse the request.** Identify Li, the Suzhou employer, the agreement and departure dates, and the reporting date. Plan sections on the persons covered, the restriction's duration and scope, and compensation.
2. **Form queries.** Search for combinations of non-compete agreements, covered employees, duration, and compensation. Expand the terms and add Suzhou or Jiangsu to retrieve material associated with the location.
3. **Retrieve materials.** Collect provisions about signed agreements, the two-year limit, and monthly payments. The candidates include Article 6 of the 2013 Labour Dispute Interpretation IV, whose 30% passage remains accessible on the SPC website [191].
4. **Rank and organize.** Deduplicate and rank the passages, retain their publication dates, and check that quotations match the source text. The historical passage is kept as relevant material without following its subsequent repeal and replacement.
5. **Repair retrieval problems.** Retry inaccessible pages and search again for any topic lacking results. The report now has authentic sources for the requested topics, so the workflow proceeds to synthesis.

6. **Produce the report.** Combine the agreement, the duration limit, and the 30% passage into the illustrative report below. Check that the quotation is accurate, the links resolve, and 30% of RMB 12,000 is RMB 3,600. These technical checks pass while the provision's temporal, territorial, and substantive conditions remain unexamined.

Report: General Workflow

Li signed a confidentiality and non-compete agreement. Its one-year duration is within the two-year limit. Article 6 of Labour Dispute Interpretation IV provides for monthly compensation at 30% of the preceding 12 months' average wage, so RMB 3,600 meets the required standard. Li must comply with the nationwide restriction on joining an artificial-intelligence business.

The analytical gap lies between finding a relevant provision and establishing that it supports a conclusion about these facts. The second workflow examines that connection before accepting the report.

Applying the workflow with legal expertise. The nine steps below follow the second design on the same request. Each step determines which facts or provisions the next stage needs and which conclusions can be supported.

1. **Identify the legal relationship and requested determination.** Classify the relationship as employment and the disputed obligation as post-employment non-competition. The task concerns Li's eligibility, the permitted scope, and the compensation requirement. Corporate-office duties and restrictions during employment are separate questions.
2. **Build a fact record.** Fix the employer's registration, operations, and contract performance in Suzhou. Record the September agreement, October departure, and November 2025 reporting date, Li's established access to protected technical secrets, the 12-month restriction, and the expressly agreed payment of RMB 3,600. Mark the prospective employer's products and competitive relationship as missing facts.
3. **Decompose the issues.** Under Articles 23 and 24 of the Labour Contract Law, distinguish the persons covered, the business and territorial scope, the maximum duration, and post-employment compensation. Article 24 identifies senior managers, senior technical personnel, and other personnel with confidentiality obligations, and sets a two-year ceiling [192]. The one-year term answers the duration question while leaving the scope and payment questions open.
4. **Retrieve by issue and source type.** Collect the Labour Contract Law, the SPC's 2020 and 2025 interpretations, and the Jiangsu provisions. Keep the historical interpretation and the Shenzhen technical-secret provisions in the candidate record so that their versions and territorial conditions can be checked. Search for rules governing omissions, exceptions, and limits alongside the affirmative provisions.
5. **Verify versions and authority.** Trace Article 6 of the 2013 Interpretation IV to the SPC's 2020 repeal decision, whose annex item 65 expressly lists that interpretation and whose effective date is 1 January 2021 [191, 193]. Retrieve Article 36 of the 2020 interpretation as the current counterpart of the 30% passage. Read it alongside the 2025 interpretation, effective 1 September 2025. Article 21 of the latter repeals a specified paragraph of the earlier interpretation and gives the new interpretation priority where earlier interpretations conflict [190, 194].
6. **Check applicability to each issue.** Article 13(1) of the 2025 interpretation supports a request to confirm that a clause has not taken effect when the employee has not known or

accessed the employer's trade secrets or intellectual-property-related confidential matters. Li's access is established in the supplied facts, so that premise is not met. Article 14 concerns restrictions during employment and does not support non-payment after departure [190]. Article 36 of the 2020 interpretation addresses omitted compensation where the worker has performed the obligation, using 30% of the preceding 12 months' average wage with the contract-performance location's minimum wage as a floor. Li already has an express compensation term [194]. Article 24 of the 2019 amended Shenzhen regulations specifies one-half of the preceding 12 months' average monthly wage, but Article 2 confines those regulations to enterprise technical-secret protection within the special economic zone. That territorial condition excludes the provision as a payment basis in Suzhou [195].

7. **Investigate facts required by the rule.** Apply Article 13(2)'s requirement that the business, geographical, and temporal scope correspond to the protected matters known or accessed; it provides for invalidity of the portion exceeding a reasonable proportion [190]. Request the prospective employer's specific products and their competitive relationship with the protected image-algorithm product. Investigate how Li's knowledge could justify a restriction extending to every artificial-intelligence business nationwide. The scope judgment remains pending that evidence.

8. **Resolve apparent conflicts.** Use Articles 2 and 28 of the Jiangsu Labour Contract Regulations for the local relationship. Article 28 sets monthly compensation at no less than one-third of the preceding 12 months' average monthly wage [196]. The verified RMB 12,000 wage gives a floor of RMB 4,000, which exceeds the agreed RMB 3,600 by RMB 400. The 30% default for omitted compensation and the Shenzhen one-half provision do not establish a different standard for these facts. The workflow resolves their different conditions before comparing amounts.

9. **Deliver and review the reasoning.** Produce the report below, accompanied by provision-to-fact links, exclusion reasons, and the outstanding questions about competing products and restrictive scope. Review whether each conclusion follows from an applicable provision and whether each unresolved judgment has a specific next step.

Report: Legal Expertise

Li's personal eligibility and the clause's scope should be examined under Articles 23 and 24 of the Labour Contract Law and Article 13 of the SPC's 2025 interpretation. Li's access to protected technical secrets is established in the supplied facts. Assessing the nationwide restriction across artificial-intelligence businesses requires the prospective employer's product information and a justification linking the restricted activities and territory to those secrets. Jiangsu Article 28 sets the monthly compensation floor at one-third of the verified RMB 12,000 wage, or RMB 4,000. The agreed RMB 3,600 falls below that floor. The 30% rule for omitted compensation does not establish compliance with this requirement, and the Shenzhen provision is outside the applicable territory.

Where the Skill designs differ. The general workflow retrieves relevant provisions and checks coverage and citation accuracy. The legally informed workflow also tests each provision's conditions against the case facts. In this example, those checks determine which version of the 30% rule to consider, which local compensation standard applies, and what product information is needed to assess scope. They also direct recovery through replacement of repealed provisions, exclusion of rules for other territories, and targeted factual inquiries. Legal expertise thus shapes both what the system retrieves and how it establishes support for its conclusions.